Patient-Level Differential Privacy in Hierarchical Federated Learning for Multimodal Medical IoT

Samartha H V
Manipal Institute of Technology Bengaluru
Manipal Academy of Higher Education
Manipal, India
samartha1.mitblr2025@learner.manipal.edu

Shreyas J
Manipal Institute of Technology Bengaluru
Manipal Academy of Higher Education
Manipal, India
shreyas.j@manipal.edu

Udayaprasad P K
Manipal Institute of Technology Bengaluru
Manipal Academy of Higher Education
Manipal, India
udayaprasad.mitblr2024@learner.manipal.edu

Shreelakshmi Yadav N
Manipal Institute of Technology Bengaluru
Manipal Academy of Higher Education
Manipal, India
shreelakshmi.mitblr2024@learner.manipal.edu

Sowmya T Rao
Manipal Institute of Technology Bengaluru
Manipal Academy of Higher Education
Manipal, India
sowmya.mitblr2024@learner.manipal.edu

***Abstract*—Hierarchical federated learning keeps Internet-of-Things data local, but its differential-privacy guarantee protects single records, whereas clinical deployments must protect patients, each with many records. This paper proposes PP-HFM, a client–edge–cloud framework whose privacy unit is the patient. Each client clips a patient's summed gradient, calibrates noise to its population and accounts for every hierarchical release. The guarantee is proved for the training transcript and holds end to end with public encoders and standardization constants (eICU, individuals at one hospital). PP-HFM is evaluated on VitalDB, an eICU cohort of 187 hospitals and, at record level, a wearable corpus. At a training budget of at most $\varepsilon = 0.9$ (1.0 with an evaluated conformal release; 0.766 at record level on WEAR), it reduces uplink traffic by 90.8–$96.1\,\%$ and modeled energy per round by 72.3–$79.6\,\%$; 95th-percentile inference latency on a workstation CPU is at most 2.3 ms. On VitalDB, partitioned by person, the guarantee costs no utility (area under the ROC curve, AUC, 0.832 against 0.820 non-private); on eICU it costs 3.1 points (inconclusive against the two-point target) and 3.9 end to end, which misses it; non-private HED-FL is more accurate (0.825 against 0.794). PP-HFM uploads 2.2–4.0 times less than re-implemented HED-FL and FL2DP and shows lower measured leakage than both in the VitalDB audit. On VitalDB and WEAR, pretrained initialization, not head-only noise, makes a strict budget survivable; the cost of the patient unit follows $A(1 - 1/k)$ in records per patient k. A VitalDB membership-inference audit with a positive control finds record-level training above its difficulty floor (likelihood-ratio attack AUC 0.576) and patient-level training at it (0.503).**

***Index Terms*—Hierarchical federated learning, Internet of Things, differential privacy, user-level privacy, membership inference, multimodal learning, edge computing, quality of service, conformal prediction, clinical decision support.**

I. Introduction

Internet-of-Things deployments increasingly gain their value from combining different kinds of sensing streams. An operating room records continuous physiological waveforms together with the structured clinical history of the patient. A wearable installation records inertial motion together with egocentric video. In both cases the streams complement each other, and a model that sees both can recognize states that neither reveals alone. The streams are also produced where the sensors are, which is rarely where computation is convenient. This work addresses the tension between where data arises and where models are trained.

The obvious solution is to send the raw streams to a central service, but this is often not possible. The data volume is large and the uplink is the scarce resource, so centralization increases both latency and energy in exactly the deployments that can least afford them. More importantly, in the settings where multimodal sensing is most valuable, the data cannot legally move at all. Clinical records are governed by consent and by data-protection laws that prohibit pooling across institutions. A model that requires centralization is therefore not merely expensive; it cannot be deployed. Federated learning [1] solves this by exchanging model updates instead of records. Hierarchical variants [2], [3] add an edge tier, so that the wide-area link is used only once every several local rounds.

Moving the model instead of the data does not, by itself, protect the data. The updates a client sends are computed from its records, and membership inference against released models is well established [4]. A formal guarantee is therefore needed. Differentially private stochastic gradient descent [5] provides one by bounding the influence of each training record and adding noise to the aggregate. However, the guarantee is stated over the unit that is clipped, and in a sensing deployment that unit is a fixed-length window, not a person. Every patient, and every participant in a continuous recording, contributes many windows. A record-level budget answers whether a particular window was used in training, but a clinical deployment must answer whether a particular *person* was. Group privacy [6] can relate the two. At the group sizes found in these datasets, however, the conversion is prohibitive: a nominal budget of one becomes a budget that supports no meaningful claim.

Existing systems have developed largely along two separate lines. Hierarchical designs make a convincing case for communication and energy savings. HED-FL [3] extends deployment lifetime by avoiding long-range transmission, HCEF [7] adapts computation and compression to device heterogeneity, and assignment-oriented work [8] reduces divergence under non-identical data distributions. None of them provides a formal privacy guarantee. Private designs do provide one: FL2DP [9] combines noise with shuffling, and RoPPFL [10] pairs local privacy with robust aggregation at the edge. Neither

is hierarchical in the sense that matters here, and neither examines how an intermediate tier changes how often each client's budget is spent. In both lines the local objective is treated as the aggregation of generic parameter vectors. Neither the presence of several modalities per record nor the grouping of records by individual plays any role. More recent systems join a hierarchy with privacy [11], [12] or privacy with multimodal clinical data [13], [14], but, to our knowledge, none states its guarantee over an individual whose records are grouped before clipping, and none accounts for that unit across an edge tier.

The review in Section II identifies five gaps that affect deployability as much as privacy.

1) *Hierarchy and privacy are designed separately.* Hierarchical systems provide no guarantee, and private systems do not examine how an intermediate tier changes each client's budget expenditure.
2) *The cost of the unit is unmeasured.* The unit at which a guarantee is stated is rarely examined, so its cost on genuinely paired multimodal data has not been quantified.
3) *A validity condition is left implicit.* Parameter-efficient private training is valid only if the corpus used to pretrain the frozen representation is disjoint from the protected population [15], [16]. This condition is often left implicit, and where it is respected, its cost is not measured.
4) *Guarantees are not audited at the right unit.* Guarantees are usually reported from the parameters of the mechanism rather than verified by attack. Where an audit is performed, it is done at the level of the record, which is exactly what a record-level guarantee already protects.
5) *Inference latency is not reported.* Federated designs report training costs, but not the inference latency experienced at the point of care.

Two remedies follow directly. First, the unit can be fixed at the point of clipping instead of being recovered afterwards through group privacy. If the gradients of an individual's records are summed before the bound is applied, that individual's influence on the released model is bounded no matter how many records they contributed. Accounting at the same level then yields a budget that is, by construction, a statement about people. Second, the resulting guarantee can be audited by membership inference at the level of the individual. For a negative finding to be informative, the attack must be strong enough: it must be shown to find leakage on a configuration that is known to memorize.

Accordingly, this work presents PP-HFM, a three-tier federated framework for multimodal sensing in which the privacy unit is the individual by construction. Records are grouped by individual before they are bounded. The noise scale is calibrated to meet the budget for every client at once, not for a single representative client, so the reported figure reflects the effect of the edge tier. Representation learning is separated from private adaptation: the encoders are pretrained on a population disjoint from every client and then frozen, so only a compact fusion head is privatized. The framework is evaluated on two corpora whose modalities are linked by the recording apparatus rather than by a pairing rule, and its central privacy result is tested on an external clinical corpus of 187 hospitals. The protocol measures communication, latency, reliability and energy on a simulated topology, and it audits the privacy guarantee by attack.

The main contributions of this paper are as follows.

- *An individual-level guarantee across the hierarchy (gap 1).* The framework enforces an individual-level guarantee within a hierarchical multimodal system and reports a budget that includes the edge tier. The accounting this requires is specified in enough detail to be checked by others: the noise is calibrated against every client rather than a representative one, and the noise sum is normalized by a public constant rather than the realized batch. An error in either is not detectable from a training log.
- *The measured cost of the unit and of disjointness (gaps 2 and 3).* The cost of the choice of unit is measured on genuinely paired clinical multimodal data. So is the cost of the disjointness condition that makes the guarantee valid end to end. Neither is assumed to be negligible.
- *A patient-level audit with a positive control (gap 4).* The guarantee is audited by patient-level membership inference, using both a threshold attack and a likelihood-ratio attack [17]. Each attack is reported together with a positive control on a configuration known to memorize. On VitalDB, record-level training remains detectably above its difficulty floor under offline LiRA, whereas patient-level training does not. The same audit, data, encoders and seeds are applied to two re-implemented prior systems, HED-FL [3] and FL2DP [9], so the comparison covers leakage as well as utility and traffic.
- *System validation on three deployments (gaps 1 and 5).* The system-level behavior is validated on three deployments that differ in topology and payload, up to 187 clients, where contention keeps update delivery below its target until the clients are spread over eight edge aggregators. Every network quantity is measured on rounds in which the payload actually crossed the simulated topology, not computed analytically. The same protocol shows that the guarantees are affordable in real time. The private mechanism takes no part of the inference path, because its noise is added to gradients during training and adds no operation at inference.
- *A scaling law for the price of the unit (gap 2).* The price of the individual-level guarantee is shown to be not a fixed quantity but a function of how much each individual contributes. With the per-step budget and the number of units per step held constant, the penalty is indistinguishable from zero when each individual contributes a single record. It grows monotonically as that number rises and saturates by eight records. At one record per individual the two units coincide by construction, so this configuration serves as a control, and the divergence at larger contributions is attributable to the unit rather than to the experimental setup.

PP-HFM builds on user-level differential privacy [18] and

parameter-efficient differentially private fine-tuning [15], [16]. The contributions are the system that composes them across every tier of a hierarchy, the proof of its patient-level guarantee, and its evaluation and audit on three multimodal corpora.

The rest of the paper is organized as follows. Section II reviews related work on hierarchical and private federated learning, on the unit at which a guarantee is stated, and on membership inference as an auditing tool. Section III states the problem and the research objectives and presents the proposed framework. Section IV describes the corpora, the implementation, the evaluation measures and the experimental protocol. Section V reports the results, Section VI discusses their interpretation and the limitations of the study, and Section VII concludes with directions for future work.

II. Related Work

A. Federated Learning and Hierarchical Aggregation

Federated learning trains a shared model by exchanging model updates instead of data [1]. In its flat form, every client communicates with a central server in every round, so the wide-area link, rather than local computation, limits how often the global model can be updated. Client–edge–cloud designs [2] address this with an intermediate tier that performs several rounds of local aggregation before each round of central aggregation. HED-FL [3] applies this principle to constrained deployments, clustering devices dynamically at the edge and extending the operational lifetime of an installation by avoiding long-range transmission. HCEF [7] adds heterogeneity awareness, adjusting per-device computation frequency and compression ratio online so that the slowest participant does not determine the round duration. Mhaisen *et al.* [8] optimize the assignment of clients to aggregators and show that an assignment based on the statistics of local data reduces model divergence under non-identically distributed data. These works establish the communication and energy benefits of hierarchy, which the present study adopts. None of them, however, provides a formal privacy guarantee. HED-FL provides none, and the assignment strategy of [8] requires clients to disclose local label statistics, which is itself a release of information about the protected population.

B. Differential Privacy in Federated Learning

Differentially private stochastic gradient descent [5] bounds the influence of each training record by clipping its gradient to a fixed norm and adding calibrated noise to the aggregate. The accumulated privacy cost is tracked by a moments or Rényi accountant [19] within the framework of [6]. In federated learning, the mechanism has been developed in two main directions. FL2DP [9] combines an exponential noise mechanism with a shuffler that removes the association between a client and its update, improving accuracy at an equivalent guarantee. RoPPFL [10] combines local differential privacy with similarity-based robust aggregation at the edge, so that the intermediate tier also verifies updates. Both show that a formal guarantee is attainable in federated learning at acceptable cost. Neither, however, accounts for the edge tier: in FL2DP the edge plays no role in the privacy accounting, and neither work examines how an intermediate tier changes the frequency with which each client's budget is spent, which determines the calibration of the noise.

C. The Unit to Which a Guarantee Applies

The distinction between record-level and individual-level guarantees is well established. McMahan *et al.* [18] introduced user-level differential privacy for federated language models, bounding the contribution of each user rather than each example, and group privacy provides the standard conversion between the two [6]. Recent work brings privacy into hierarchical and multimodal settings. PCRHFeL [11] chains sparsification, Wasserstein differential privacy and robust aggregation in a client–edge–cloud hierarchy. Fulcrum [12] allocates noise per participant according to its exposure within a regional aggregate, a calibration close in spirit to the per-client noise multipliers used here but defined for a participant's class distribution rather than an individual's records. Multimodal clinical systems add noise to the local updates of an EHR–ECG fusion model [13] or give client-level guarantees for multimodal medical imaging [14]. To our knowledge, none of these groups an individual's records before clipping or measures the cost of that unit. What is missing is a measurement of the cost of the individual-level guarantee in the setting that most requires it. In clinical monitoring, every participant contributes many records, because a single episode of care yields many time-aligned windows. The conversion factor therefore applies to the entire population rather than to a few atypical participants. The literature reports neither the utility lost when the unit is moved from the record to the individual on genuinely paired multimodal clinical data, nor how that cost varies with the number of records per individual. Without the latter, a single measurement cannot be distinguished from a property of the particular cohort.

D. Parameter-Efficient Private Adaptation

The noise a parameter vector must absorb under a fixed budget grows with its dimension, so private training of a full network degrades severely at the budgets required for clinical use. Restricting private optimization to a compact trainable subset above frozen pretrained representations is an established remedy: Li *et al.* [15] and Yu *et al.* [16] show that parameter-efficient differentially private fine-tuning is considerably more robust at small ε than end-to-end private training. The present work adopts this approach and claims no novelty for it. The approach, however, is valid only if the corpus used to obtain the frozen representations is disjoint from the protected population. Otherwise, the representations are derived from that population without a guarantee, and the budget reported for the trainable component does not describe the system as a whole. This condition is often left implicit, and where it is satisfied, its cost in utility is rarely reported.

E. Membership Inference as Empirical Auditing

A differential privacy guarantee is an upper bound on the influence of a participant; it does not show what an

adversary achieves in practice. Membership inference provides this complementary measurement. Shokri *et al.* [4] introduced the shadow-model formulation, and thresholding of the loss or confidence [20] remains the most commonly reported attack. Such attacks are weak, because a single global threshold ignores the varying difficulty of individual examples. An easily fitted non-member may therefore score above a poorly fitted member, and the reported attack strength is a lower bound on leakage rather than an estimate of it. Watson *et al.* [21] showed that calibrating for per-example difficulty changes conclusions substantially. Carlini *et al.* [17] formalized this calibration as a likelihood-ratio test between per-example in- and out-distributions estimated from shadow models, and recommended evaluating attacks by their true-positive rate at a low false-positive rate. Two limitations remain. First, membership is almost always evaluated at the level of the training record, which a record-level guarantee already protects, rather than at the level of the individual. Second, negative results are rarely accompanied by evidence that the attack could produce a positive one, so the absence of detected leakage cannot be distinguished from a weak adversary.

F. Multimodal and Clinical Federated Learning

Federated learning is particularly well motivated in medicine, where data governance rather than data volume prevents pooling at a single site [22], [23]. Most federated work, including the systems discussed above [3], [7], [9], [10], treats the local objective as the aggregation of generic parameter vectors and does not model multiple modalities per record, although clinical data are typically multimodal and the available modalities may differ between institutions. In existing multimodal federated benchmarks, the modalities are often paired by a construction rule rather than by common origin. An apparent fusion benefit on such a benchmark may reflect the pairing rule rather than complementary information, so it cannot support a claim about fusion. Corpora in which both modalities come from the same individual during the same episode and are aligned at capture [24], or synchronized in hardware [25], avoid this problem, because the pairing is a property of the source.

G. Uncertainty Quantification under a Privacy Budget

Conformal prediction produces prediction sets with a distribution-free coverage guarantee and requires neither retraining nor distributional assumptions [26], [27]. This makes it suitable for clinical deployment, where a point prediction without a confidence statement is of limited use. In a private system, however, conformal calibration requires the release of a quantile of nonconformity scores computed on held-out data from the protected population, and this release is itself a disclosure. Differentially private quantile release by the exponential mechanism is standard [28]. The open issue is accounting: the cost of calibration composes with the cost of training, and a system that reports only the latter understates its total budget, by an amount that grows with the number of calibration releases.

H. Real-Time Inference and Quality of Service at the Edge

A federated model is trained once and queried continuously. After deployment, its clinical usefulness depends on the delay between the arrival of a measurement and the availability of a prediction, not on the time to convergence. The federated literature has studied the latter extensively and the former very little. HCEF [7] adapts computation frequency and compression ratio online, but it optimizes the duration of a training round. RoPPFL [10] verifies updates at the edge, prioritizing robustness over response time. HED-FL [3] minimizes the energy needed to train a model rather than the latency of using it. Inference cost has mainly been studied in the edge-computing literature rather than the federated literature [29], where reduced-precision arithmetic, in particular integer quantization, is an established way to lower memory footprint and inference time on commodity hardware [30]. A federated system may therefore meet a strict privacy budget and still be unusable at the bedside if its prediction latency exceeds the time within which a clinician can act. No existing federated system reports end-to-end inference latency under its deployed privacy mechanism together with the communication and energy costs of training the model.

I. Summary and Research Gap

Five gaps follow from this review.

1) Hierarchy and formal privacy have been developed largely separately: hierarchical systems provide no guarantee, and private systems do not examine how an intermediate tier affects budget expenditure. Their interaction is therefore unmeasured.
2) The mechanism for stating a guarantee over individuals is established, but its cost on genuinely paired clinical multimodal data is not. Without a characterization of how this cost varies with the number of records per individual, a single measurement cannot be distinguished from an artifact of the cohort.
3) Parameter-efficient private adaptation is widely used, but the disjointness of the pretraining corpus on which its validity depends is often left implicit, and its cost is not reported.
4) Guarantees are usually reported from the mechanism's parameters rather than audited by attack. Where audits exist, they operate at the record level, and null results are rarely accompanied by evidence that the attack could detect leakage.
5) The cost of federated designs is reported in training terms, such as rounds to convergence, bytes exchanged and energy consumed, while the inference latency experienced by the clinician is not measured, although it determines whether the system can be used at the point of care.

The present work addresses these five gaps together. It combines a two-tier hierarchy with a guarantee enforced at the level of the individual at the point of clipping, rather than obtained by conversion afterwards, and calibrates the noise against every client simultaneously so that the reported budget reflects the edge tier. It measures the utility lost by

moving the unit from the record to the individual, and characterizes this cost as a function of the number of records per individual rather than at a single operating point. It enforces and verifies the disjointness of the pretraining population and reports the utility cost of an end-to-end valid guarantee. It audits the guarantee by individual-level membership inference, using both a threshold attack and a per-individual calibrated likelihood-ratio attack, each reported with a positive control on a configuration known to memorize. Finally, it reports the end-to-end prediction latency, the traffic carried by the hierarchy and the energy consumed by each tier for the same configuration that carries the privacy guarantee.

III. Methodology

A. Problem Statement

Multimodal Internet-of-Things deployments combine sensing streams of different kinds, such as physiological monitors with clinical records or inertial sensors with video, at the location where the streams are produced. The training data are therefore distributed, and in clinical settings they cannot be centralized, because consent and data-protection regulations prohibit pooling. Federated learning addresses this by moving the model instead of the data, but it does not provide a privacy guarantee, because client updates are functions of the client's records. A suitable framework must therefore combine distributed multimodal training with a formal guarantee, within the network, latency and energy constraints of an edge installation.

Differentially private federated learning provides such a guarantee, but the *unit* of the guarantee in clinical settings has received little attention. Standard differentially private stochastic gradient descent [5] clips and accounts per training record. In a monitoring dataset, a record is a fixed-length window rather than a person, and every patient in the cohort used here contributes several windows. A record-level budget therefore bounds whether a particular window was used in training, whereas a clinical guarantee must bound whether a particular *patient* was. Prior federated systems address only one side of this problem. HED-FL [3] provides a hierarchical tier without differential privacy, FL2DP [9] provides differential privacy without a hierarchy, and neither these nor the heterogeneity-aware and assignment-oriented designs [7], [8] model multiple modalities per record or the grouping of records by individual.

The problem addressed is therefore the following. Several institutions each hold multimodal recordings of their own patients, every patient contributes many records, and no record may leave its institution. A shared model must be trained through a client–edge–cloud hierarchy such that

1) the participation of any single patient, with all of their records, is protected by a formal (ε, δ) guarantee that covers every message the system releases, edge tier included;
2) the model remains close in utility to an identical non-private system; and
3) the system remains deployable on IoT infrastructure, with low uplink traffic, reliable delivery, low inference latency and low energy.

These requirements interact. Protecting patients instead of records reduces the number of protected units and therefore raises the relative noise; clients partitioned by institution differ greatly in size, so a noise level fitted to the smallest client degrades every larger one; and an edge tier changes how many optimization steps each client takes, and hence how much of its budget it spends. Table I turns the three requirements into seven quantitative targets, fixed before any experiment was run.

B. Overall Framework of the Proposed PP-HFM

This work therefore proposes a framework in which the privacy unit is the patient by construction, and in which the hierarchy and the guarantee are designed jointly. The system, PP-HFM (Patient-Private Hierarchical Federated Multimodal learning), has three tiers: institutional clients that hold patient-grouped multimodal records and never release them, edge aggregators that perform several local rounds before communicating upward, and a cloud aggregator that maintains the global model. Within each client, representation learning is separated from private adaptation. Encoders pretrained on a disjoint public population are frozen, and only a compact fusion head is trained under a differentially private objective with patient-level sensitivity. Fig. 1 shows the overall organization.

Each client samples a batch of patients rather than windows, sums the gradients of all windows of each sampled patient into one per-patient gradient, clips that gradient, and adds noise to the aggregate once per optimization step. The influence of any individual on the released model is therefore bounded regardless of the number of records contributed. The resulting update is compressed and sent to the client's edge, which averages the updates of its clients. After several edge rounds, the edges report to the cloud, which averages across edges and redistributes the global model.

The framework comprises the following components, each described in one of the following subsections.

1) *Frozen multimodal backbone* (Section III-D): fixed encoders, one per modality, and a compact trainable fusion head.
2) *Patient-level privacy mechanism* (Section III-E): enforces sensitivity with respect to the individual.
3) *Noise calibration across clients* (Section III-F): meets the budget for every client despite large differences in client size.
4) *Two-tier aggregation* (Section III-G): produces the global model while using the wide-area link once per several local rounds.
5) *Update encoding* (Section III-H): reduces the transmitted payload.
6) *Deployment path* (Section III-I): executes the released model within a latency budget.
7) *Network and energy instrumentation* (Section III-J): measures the communication and energy costs of the hierarchy.
8) *Empirical audit* (Section III-K): tests the resulting guarantee by membership inference.

The encoder architecture and the aggregation rule are adopted from established work and are not claimed as contributions.

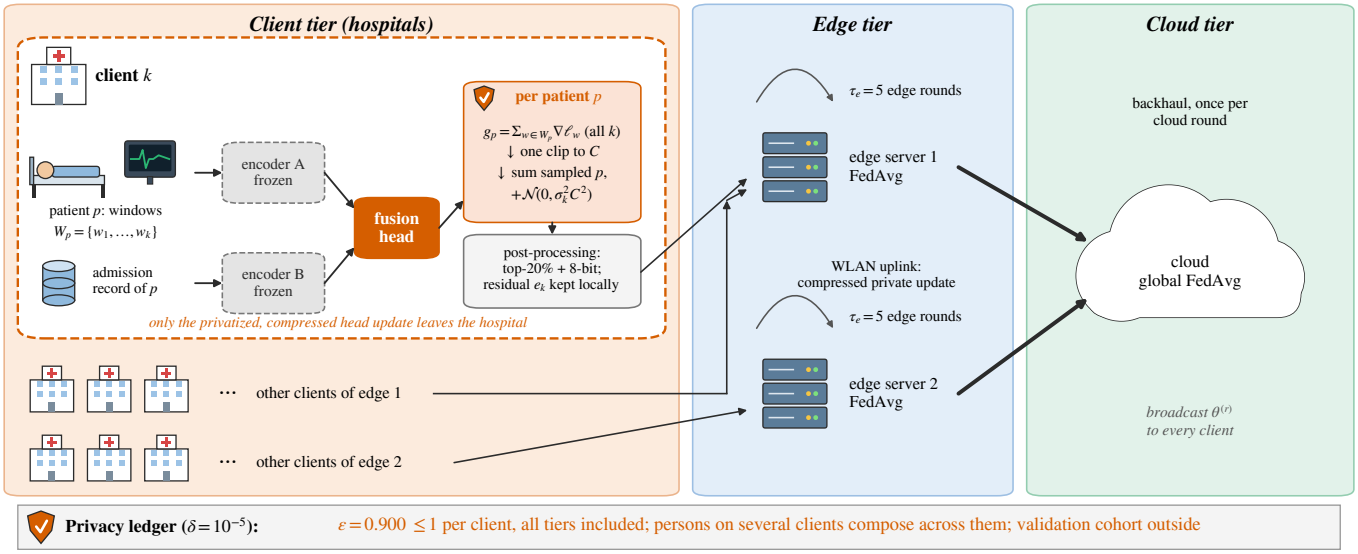


Fig. 1. Architecture of PP-HFM. Each institutional client holds patient-grouped multimodal records that never leave the institution, runs frozen pretrained encoders, and privately trains only the fusion head, with gradients summed per patient before clipping. Two edge aggregators perform τ_e rounds of weighted averaging for every cloud round, and the cloud averages the noisy, compressed updates globally.

C. Overall System Architecture

PP-HFM is a three-tier federated system built on a frozen multimodal backbone. Responsibilities are separated: the encoders provide fixed representations, a compact head contains all trainable and privatized parameters, an edge tier absorbs local rounds to reduce wide-area traffic, and a cloud tier maintains the global model. The encoder design and the weighted averaging rule are adopted from prior work. The design contribution lies in the unit at which privacy is enforced and in the accounting of the reported budget.

The architecture is shown in Fig. 1. An institutional client holds a set of patients, each contributing several time-aligned multimodal records. The records are encoded by two frozen modality-specific encoders and combined by a trainable fusion head. The head is optimized under a differentially private objective, and its compressed update is the only quantity that leaves the institution. Updates are averaged at the edge over the clients it serves. After τ_e such rounds, the edge models are averaged at the cloud and redistributed. No record, single-record gradient or uncompressed update crosses an institutional boundary.

The system consists of three functional groups. The first is the frozen backbone: two modality-specific encoders, pretrained on a population disjoint from every federated client and not adapted during federated training. The second is the privatized trainable component: the fusion head, the patient-level clipping and noise mechanism that governs its optimization, and the calibration procedure that sets the noise scale. The third comprises the communication components: the update encoding and the two-tier averaging.

Information flows in a group-then-bound structure. Within a client, records are encoded and combined by the head, and gradients are grouped *by patient* before clipping, so that the clipped quantity is a patient's total contribution rather than a single record's. Noise is added once to the aggregate of the clipped per-patient gradients, after which the update is compressed and transmitted. Communication is hierarchical: client to edge in every edge round and edge to cloud once per cloud round, with the ratio set by τ_e.

This design follows from the constraints of the setting. Grouping gradients by patient before clipping makes the guarantee a statement about individuals, and avoids the severe degradation of a group-privacy conversion. Restricting private optimization to a small head over frozen features keeps the dimension of the perturbed vector low, which is necessary for acceptable utility under a tight budget. Pretraining the features on a disjoint population keeps the guarantee valid end to end. The edge tier reduces wide-area transmissions by a factor of τ_e without changing the learning objective. Because the encoders are fixed and the aggregation rule is standard, observed differences can be attributed to the privacy unit and its accounting rather than to the representation or the averaging.

D. Frozen Multimodal Backbone

A tight privacy budget requires a large noise scale, and the noise a parameter vector must absorb grows with its dimension, so private training of a full network degrades severely. Parameter-efficient differentially private fine-tuning, which restricts private optimization to a small trainable subset over frozen pretrained features, is considerably more robust at small ε than end-to-end private training [15], [16]. PP-HFM adopts this design unchanged. The framework adds the condition under which the resulting guarantee remains valid, and a measurement of the cost of that condition.

The backbone consists of two modality-specific encoders. The temporal encoder takes a multichannel physiological time series as input. A multi-scale convolutional stem applies three parallel one-dimensional convolutions with different kernel

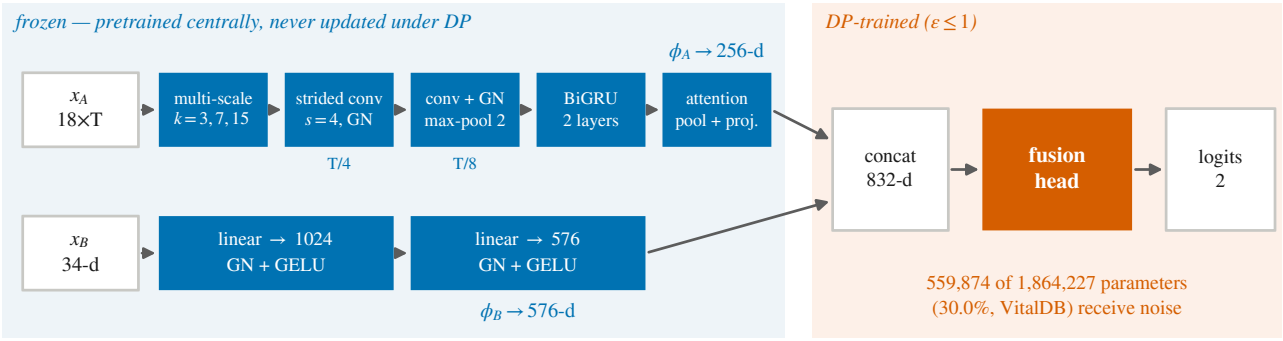


Fig. 2. Encoder–fusion backbone and the privacy boundary. Both encoders are pretrained centrally and frozen; only the fusion head is trained under differential privacy, so noise is applied to 559,874 of 1,864,227 parameters. Section III-D defines the two pretraining configurations, and Section IV-B records which produced which result. No batch normalization appears anywhere, because it mixes statistics across a batch and would make the per-patient gradient ill-defined.

widths to capture fine, intermediate and coarse temporal patterns, and concatenates their outputs. A strided convolution then reduces the temporal resolution, a further convolution with max pooling halves it again, a two-layer bidirectional recurrent layer models longer-range dependencies, and an attention-pooling layer aggregates over time with learned weights. Because pooling is adaptive, the encoder accepts sequences of any length. The static encoder processes a fixed-length vector of preoperative clinical and laboratory variables with a two-layer perceptron with normalization and a nonlinearity. The two embeddings are concatenated and mapped to class logits by the fusion head, as shown in Fig. 2.

Two properties of the backbone are required by the privacy mechanism. First, its parameters remain frozen throughout federated training, so all trainable and privatized capacity resides in the fusion head, which holds 559,874 of the 1,864,227 parameters. The private optimization is therefore close to a linear probe over fixed features. Second, every normalization layer is a group or layer normalization; no batch normalization is used. Batch normalization mixes statistics across the examples of a batch, so per-example, and therefore per-patient, gradients would not be well defined, and the privacy accounting would be invalid. The recurrent layer is likewise implemented in a form compatible with per-sample gradients.

The pretraining corpus must satisfy an additional condition. If the encoders are pretrained on data from the federated clients, they have been exposed to the protected population without a guarantee, and the budget reported for the head does not describe the system end to end. PP-HFM therefore distinguishes two encoder configurations.

In the *oracle* configuration, the encoders are pretrained on the pooled training data of the federated clients. This configuration does *not* provide a valid end-to-end guarantee and is not proposed for deployment. It is used to characterize the mechanism, because it isolates the attainable performance of the head-only recipe from the origin of the representation. In the *disjoint* configuration, a set of individuals is reserved before federated training, disjoint at the individual level from every client's training, validation and test splits, and the encoders are pretrained only on this set. Disjointness is verified explicitly. This configuration provides the guarantee claimed in this paper, and its utility cost is measured. Section IV-B specifies which configuration produced each result, and Section V-A reports the cost of moving between them.

E. Patient-Level Privacy Mechanism

A clinical system must provide a guarantee about individuals, whereas the standard mechanism provides one about records. Differentially private stochastic gradient descent bounds the influence of a training *record* by clipping its gradient to a fixed norm and adding noise to the aggregate [5]. When each record corresponds to a distinct person, the two coincide. In a monitoring dataset, however, a record is a fixed-length window and a single operation yields many windows, so removing one patient removes as many records as that patient contributed. An (ε, δ)-differentially private mechanism is $(k\varepsilon,\ ke^{(k-1)\varepsilon}\delta)$-differentially private for groups of size k [6]. Applied at the mean group size of this cohort, this conversion turns a nominal $\varepsilon = 1$ into $\varepsilon \approx 8$ with $\delta \approx 9 \times 10^{-2}$, which does not support a clinical claim. The gap between the two units is illustrated in Fig. 3.

Whether the two units differ depends on the deployment. When each client corresponds to a single individual, as in a wearable installation where one participant carries one device, client-level, user-level and individual-level guarantees coincide. When a client holds records from many individuals, as an institution holding many patients does, the units differ for every participant. The mechanism below addresses the second case. The framework applies to both, but only the second exhibits the gap, and the evaluation is organized accordingly.

The mechanism enforces the guarantee at the point of clipping rather than by conversion afterwards. The sampling unit is the patient: at each optimization step, a client samples patients independently at a fixed rate, so the rate entering the accountant is defined over patients rather than windows. For each sampled patient, the gradients of all of that patient's windows are summed into one per-patient gradient, and this sum is clipped to the clipping norm. Clipping the sum makes

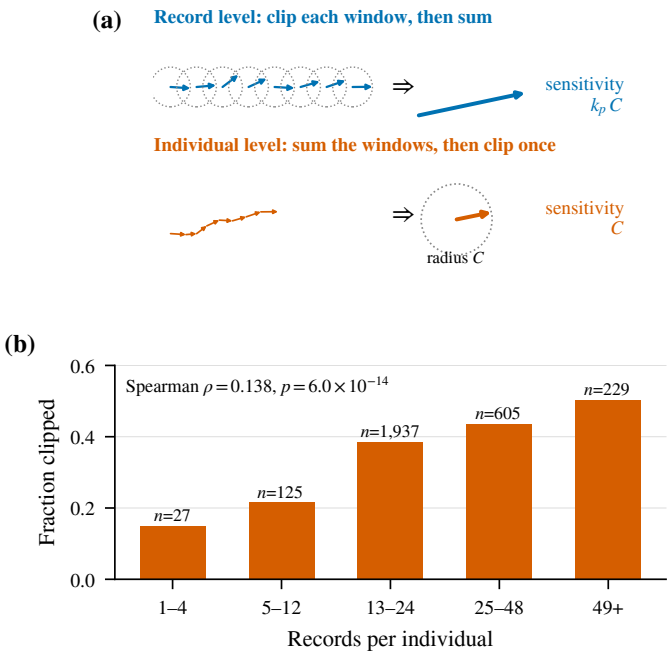


Fig. 3. Where the guarantee is decided. (a) Clipping each window to C bounds the influence of a *window*: a patient contributing k_p windows can still displace the aggregate by $k_p C$. Summing a patient's windows before clipping bounds the influence of the *patient* at C, however many windows that patient contributed. (b) On eICU at $C = 1$, the fraction of individuals whose summed gradient is clipped rises with the number of records they contribute; bar labels give individuals per bin.

the sensitivity patient-level: a patient can change the aggregate by at most the clipping norm regardless of the number of windows, whereas clipping each window separately would allow a change proportional to that number. Gaussian noise calibrated to the same norm is then added once per step to the aggregate of the clipped per-patient gradients, the result is divided by the *expected* batch size $q|\mathcal{P}_k|$, and the head parameters are updated. The divisor must not be the realized batch size $|S|$: the noise norm reveals the divisor, and $|S|$ depends on whether a given patient is present, so normalizing by it releases information the accountant does not cover. For the same reason a step whose sample is empty still releases the noise. The mechanism is stated in Algorithm 1. For compute, at most 64 records of a patient enter its summed gradient; the rule is fixed and public, so the guarantee is unaffected, and it truncates $4.1\,\%$ of eICU patients ($3.5\,\%$ of training windows) and no VitalDB patient.

a) Objective solved: Without clipping, the expected step of client k is $|\mathcal{P}_k|^{-1} \sum_p g_p = \bar{k}_k \nabla F_k(\theta)$, where F_k is the record mean of the client's empirical risk and $\bar{k}_k = n_k/|\mathcal{P}_k|$ the mean number of records per patient. A local pass takes $|\mathcal{P}_k|/B$ such steps, $\bar{k}_k$ times fewer than a record-level pass with B records per step, so to first order in the learning rate both displace client k by $\eta\,(n_k/B)\,\nabla F_k$. Before clipping, the patient-level mechanism therefore targets the declared record-mean objective with unchanged client weights, in fewer and larger steps. The argument is for the plain gradient step of Algorithm 1; the implementation passes the privatized gradient to Adam, which is post-processing and leaves the guarantee

Algorithm 1 Patient-level private local update at client k

Require: frozen θ_{enc}; head θ_{head}; patients $\mathcal{P}_k$ with records $\{W_p\}$; clip C; noise σ; rate q from (3); step size η

for each local step **do**
  Sample $S \subseteq \mathcal{P}_k$, each p independently w.p. q
  for $p \in S$ **do**
    $g_p \leftarrow \sum_{(x,y)\in W_p} \nabla_{\theta_{\text{head}}} \ell(f_\theta(x), y)$ ▷ group by patient
    $\bar{g}_p \leftarrow g_p \cdot \min(1, C/\|g_p\|_2)$ ▷ bound the sum
  end for
  $\tilde{g} \leftarrow \frac{1}{q\,|\mathcal{P}_k|}\big(\sum_{p\in S} \bar{g}_p + \mathcal{N}(0, \sigma^2 C^2 I)\big)$ ▷ also when $S = \emptyset$
  $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \eta\,\tilde{g}$
  Accountant.step(σ, q)
end for
return θ_{head}

unchanged but rescales coordinates adaptively, so the argument holds only approximately. Clipping is what changes the objective: each patient's summed gradient is bounded as a whole, the surrogate of user-level DP-SGD [18], which down-weights patients whose summed gradient exceeds C, typically those with many records. This change belongs to the unit and is part of its measured cost.

Accounting uses the same patient-level sampling rate, so the reported budget is patient-level by construction. A single accountant instance persists for the whole of a client's participation, so composition across rounds is preserved. The construction has two costs, which this work measures: the number of effective units per step decreases from windows to patients, and a patient with many windows has their summed gradient clipped more strongly than a patient with few. The dependence of this cost on the number of records per individual is examined empirically by holding the first effect fixed and varying the second (Section V-B3).

F. Noise Calibration Across Heterogeneous Clients

The noise multiplier determines the budget for a given clipping norm, and depends on the sampling rate and on the number of optimization steps of a client. In a federated system partitioned by institution or specialty, these quantities differ substantially between clients: a small client samples a large fraction of its population per step, whereas a large client samples a small one. A single noise multiplier calibrated for the most demanding client satisfies the budget everywhere, but imposes that client's noise on every other; on the external corpus, a hospital of fifteen patients sampled at rate one sets the multiplier for all 187 clients.

When each patient is held by exactly one client, as (A1) requires, the clients' mechanisms act on disjoint sets of individuals, and by parallel composition the guarantee for a patient is that of the client that holds them; individuals held by several clients are treated where the scope of the theorem is stated. Each client therefore receives its own noise multiplier σ_k, the smallest that meets the training budget for that client's sampling rate and step count, Eq. (6). The multiplier depends

Algorithm 2 PP-HFM cloud round r

Require: global model $\theta^{(r-1)}$; edge map $e(\cdot)$; edge rounds τ_e

$\theta_e^{(r,0)} \leftarrow \theta^{(r-1)}$ for all edges e
for $s = 1$ **to** τ_e **do**
  for each client k **in parallel do**
    $\theta_k^{(r,s)} \leftarrow$ Algorithm 1 from $\theta_{e(k)}^{(r,s-1)}$
    Transmit $Q\big(\text{top-}\kappa(\theta_k^{(r,s)} - \theta_{e(k)}^{(r,s-1)})\big)$
  end for
  Each edge aggregates by (8)
end for
Cloud aggregates by (9) and broadcasts $\theta^{(r)}$
return $\theta^{(r)}$

only on the public client size, so it releases nothing further. Accounting uses the privacy loss distribution accountant [31], which gives a tighter upper bound than Rényi accounting for the same mechanism. The comparisons between arms in Section V-B use the single most conservative multiplier, the maximum of σ_k, with Rényi accounting, which holds every arm to the same noise.

Two details are essential. First, the number of optimization steps entering the calibration must be the number the client actually executes; a count that is too low underestimates the required noise. Second, the population and sampling rate of each client are read from the partition files used in training rather than from cached metadata, because metadata that no longer matches the data produces a budget that is violated without any warning. Both details are stated as part of the method because violations of either are not detectable from training logs and can only be found by independent recomputation.

G. Two-Tier Hierarchical Aggregation

In a flat federated system, every client update is sent to the cloud in every round, so the wide-area link limits the update frequency. An intermediate aggregation tier removes this limitation [2], [3]: several rounds of local optimization and edge averaging are performed for each round of cloud averaging, so the wide-area link is used once per group of local rounds. PP-HFM adopts this organization, with τ_e edge rounds per cloud round, and claims no novelty for the averaging rule.

Aggregation is performed in two stages, each weighted by the fixed number of training records of the participants. The weights are public constants: a weight computed from the records drawn in a given round would depend on which patients were sampled and would reach the edge unprotected. Within a cloud round, each client initializes from its edge model, performs one private local pass, and sends its encoded update. The edge computes the weighted average over its clients and returns it, and this is repeated τ_e times. The edges then report to the cloud, which computes the weighted average over edges and broadcasts it as the global model for the next round. Non-floating-point parameters are passed through unchanged. The procedure is given in Algorithm 2.

The parameter τ_e controls the trade-off of the hierarchy. With $\tau_e = 1$, the edge tier is merged into the cloud tier, which recovers conventional single-tier federated averaging; this configuration is used to isolate the contribution of the hierarchy. Larger values reduce wide-area transmissions proportionally but allow edge models to diverge further between synchronizations. Because the privacy mechanism operates entirely within a client and aggregation uses only privatized updates, τ_e affects communication and convergence, and enters the guarantee only through the number of optimization steps, which the calibration already accounts for.

H. Update Encoding

A further component is adopted from established practice and is not claimed as a contribution. It reduces the transmitted payload. Each client update is sparsified by retaining only its largest-magnitude coordinates, and the retained values are uniformly quantized to a reduced bit width. The residual removed by sparsification is stored locally and added to the next round's update, so suppressed coordinates are transmitted later rather than lost. The encoding is applied after the privacy mechanism and is therefore post-processing; by the post-processing property of differential privacy, it does not affect the guarantee and can be tuned for bandwidth alone.

A private conformal threshold, released per client by the exponential mechanism over ranks [28], [32] at $\varepsilon_{\text{conf}} = 0.1$ with one score per calibration patient, was also evaluated. At about a hundred calibration patients per client it does not attain coverage (Section V), so it is not part of the system, and the reported budget is the training budget alone.

I. Deployment Path and Inference Under a Latency Budget

The clinical value of the trained model depends on the delay between the availability of a measurement and the return of a prediction, which is a property of the deployment path. This path is therefore specified as part of the system. The trained network is exported to a portable computation graph and executed at full precision. Post-training eight-bit quantization [30] was evaluated as an alternative and rejected, because on a CPU provider it increased rather than reduced latency (Section V). Any transformation of the released model after federated training is post-processing and does not affect the privacy guarantee. Inference is executed at the edge tier rather than in the cloud, so the wide-area link is not on the critical path.

The delay is decomposed into four separately measured stages, because an end-to-end figure within budget may still be dominated by a removable stage. Preprocessing constructs and normalizes the input tensors from the sensor stream; the forward pass evaluates the exported graph; the threshold step of the evaluated conformal set, which is not deployed, is kept as a conservative upper bound; and a final stage converts the output into the format required by the consuming system.

Latency is characterized by its distribution rather than its mean, and the budget must be met at the 95th percentile, since a clinical service level is defined by its tail. The measurement configuration is fixed in advance: the full-precision graph is executed on a general-purpose CPU provider, representing an edge node without an accelerator, for one thousand timed runs

TABLE I
DESIGN OBJECTIVES AND THEIR PRE-REGISTERED TARGETS. THE PROBLEM SPECIFICATION FIXES A NUMERIC THRESHOLD FOR COMMUNICATION REDUCTION BUT ASKS ONLY THAT THE LATENCY BENEFIT OF THE HIERARCHY BE QUANTIFIED, SO O1B IS STATED AS A DIRECTION RATHER THAN A PERCENTAGE. MEASURED VALUES ARE REPORTED IN SECTION V.

	Quantity	Target
O1	Uplink traffic reduction vs. centralized	$\geq 50\%$
O1b	Training-round latency vs. centralized	improvement
O2a	Privacy budget, composed total	$\varepsilon \leq 1$
O2b	Accuracy loss under the budget	≤ 2 pp
O3a	Inference latency, 95th percentile	< 100 ms
O3b	Successful model updates	$> 99\%$
O4	Energy saving per round vs. centralized	$\geq 20\%$

after one hundred discarded warm-up runs, and each of the four stages is timed within every run. A CPU provider is used because an edge deployment cannot be assumed to have a GPU.

J. Network and Energy Instrumentation

Three of the four design objectives concern quantities that cannot be observed from training: the wide-area traffic of the hierarchy, the reliability of update delivery, and the energy consumed per round by each tier. These are properties of the deployed topology and are therefore measured on simulated models of that topology rather than estimated from parameter counts.

The network is modeled with the three tiers of Section III-C. The device-to-edge segment uses a wireless local-area link and the edge-to-cloud segment a point-to-point backhaul, so the asymmetry between frequent local exchange over a low-cost link and infrequent global exchange over a high-cost link is represented explicitly. Each round is simulated in both directions, the downlink broadcast of the global model and the uplink transmission of client updates. Traffic volume, delivery reliability and end-to-end delay are recorded per round for the proposed configuration and for the flat and centralized baselines. Per-round delay quantifies the latency benefit of the hierarchy during training, which is distinct from the inference latency of the deployment path in Section III-I. Because the payload is the compressed update of Eqs. (10)–(11), the measured traffic is that of the specified system rather than of an uncompressed approximation.

Energy is modeled per tier from the power characteristics of the hardware and the completion time of the assigned work. The device tier represents constrained embedded nodes, the edge tier server-class accelerated hardware, and the cloud tier the aggregation host. Energy is reported per tier because a hierarchical design redistributes work rather than reducing it, and an aggregate figure would not show whether a saving at one tier is offset by an increase at another.

The targets for these quantities were fixed in advance and are listed in Table I. The measured values are reported in Section V.

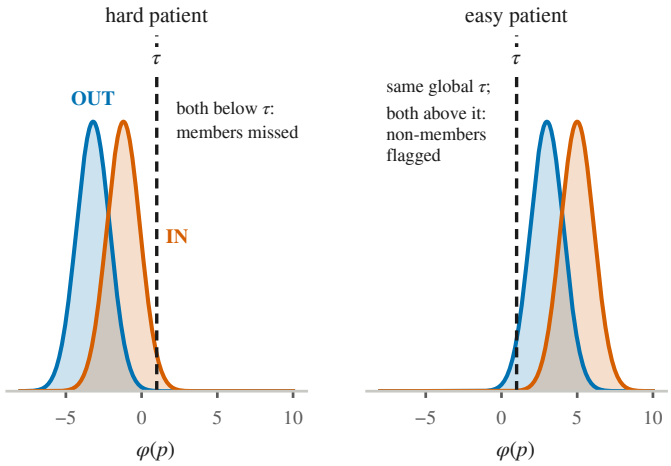


Fig. 4. Why per-patient calibration is necessary. Under a single global threshold an easy non-member can outscore a hard member, so membership signal is buried in per-patient difficulty. Fitting each patient's own in- and out-distributions from shadow models removes that confound and scores membership as a likelihood ratio.

K. Empirical Auditing of the Guarantee

A differential privacy guarantee is an upper bound on the influence of an individual. It does not show what an adversary achieves in practice, nor that a system with a weaker guarantee is exploitable. Membership inference provides this complementary measurement [4]. PP-HFM is audited by individual-level membership inference. Members are the patients in the clients' training splits and non-members are patients held out entirely. Both groups are scored by the same released model using the same procedure, so any separation between them reflects membership.

Two attacks of different strength are used. The first compares each patient's mean per-record loss with a global threshold [20]. Aggregation to the patient level before thresholding is essential: a per-record attack measures record membership, which a record-level guarantee already protects, whereas averaging over a patient's records converts weak per-record signals into a per-individual signal. The second is a likelihood-ratio attack [17] adapted so that the unit of membership is the patient. Shadow models are trained on independently sampled halves of the patient population, with all records of a patient assigned together, and each shadow is trained with the same recipe as the target model. For each patient, the distribution of a logit-scaled confidence statistic is estimated separately over the shadows that include and exclude that patient, and membership is scored by the ratio of the two densities. A global threshold is confounded by per-patient difficulty, so the attack strength it reports is a lower bound on leakage rather than an estimate [21]. Per-patient calibration removes this confound, as illustrated in Fig. 4.

Attacks are evaluated by the area under the receiver operating characteristic curve and by the true-positive rate at low false-positive rates, because an attack close to chance on average may still identify a few individuals with high confidence. When the number of non-members is too small to resolve a given false-positive rate, the value is reported as

TABLE II
NOTATION.

Symbol	Meaning
K, E	number of clients, of edge aggregators
$\mathcal{P}_k$, W_p	patients at client k; records of patient p
$k_p = \lvert W_p \rvert$	records contributed by patient p
R, τ_e	cloud rounds; edge rounds per cloud round
B	patients sampled per optimization step
C, σ	clipping norm; noise multiplier
q	Poisson sampling rate over patients
(ε, δ)	privacy budget
θ_{enc}, θ_{head}	encoder, fusion-head parameters

undefined rather than interpolated.

A negative audit result is informative only if the attack is capable of detecting leakage. The same procedure is therefore applied to a configuration known to memorize, namely non-private training of the same head over the same frozen encoders, and is required to detect clearly elevated leakage there. This positive control is a precondition of the audit and is reported with every result it qualifies.

L. Training Objective and Privacy Accounting

Let K clients be indexed by k, with client k holding a local dataset $\mathcal{D}_k$ of n_k records and $n = \sum_k n_k$. Clients are assigned to edges by a fixed map $e(\cdot)$. The records of client k are partitioned by individual into a set of patients $\mathcal{P}_k$, and W_p denotes the records contributed by patient p, with $k_p = |W_p|$; the partition is constructed so that W_p never spans two clients or two evaluation splits, which holds by construction on eICU and on the person-level VitalDB partition (Section IV). Model parameters are written $\theta = (\theta_{\mathrm{enc}}, \theta_{\mathrm{head}})$, and ℓ is the cross-entropy; the loss is unweighted, because class weights computed from the training labels would be a further unprotected release. The notation is summarized in Table II.

The learning objective is the standard weighted empirical risk over clients,

$$\begin{aligned} \min_{\theta}\ F(\theta) &= \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \\ F_k(\theta) &= \frac{1}{n_k} \sum_{(x,y)\in\mathcal{D}_k} \ell\big(f_\theta(x), y\big), \end{aligned} \tag{1}$$

minimized only over θ_{head}, since θ_{enc} is pretrained on a disjoint public population and held fixed,

$$\theta_{\mathrm{enc}} \text{ fixed}, \qquad \theta_{\mathrm{head}} \leftarrow \text{privately optimized}. \tag{2}$$

At each optimization step client k draws a subset $S \subseteq \mathcal{P}_k$ by including each patient independently with probability

$$q_k = \min\Big(1,\ \frac{B}{|\mathcal{P}_k|}\Big), \tag{3}$$

a rate defined over patients rather than records, which makes the accounted quantity patient-level. For each sampled patient the record gradients are accumulated and the sum is bounded,

$$\begin{aligned} g_p &= \sum_{(x,y)\in W_p} \nabla_{\theta_{\mathrm{head}}}\, \ell\big(f_\theta(x), y\big), \\ \bar{g}_p &= g_p \cdot \min\!\left(1, \frac{C}{\|g_p\|_2}\right). \end{aligned} \tag{4}$$

Equation (4) determines the unit of the guarantee: because the bound is applied to the sum, the influence of patient p is at most C irrespective of k_p, whereas bounding each record separately would admit an influence of $k_p C$. A single perturbation scaled to the same norm is then applied to the aggregate,

$$\tilde{g} = \frac{1}{q_k |\mathcal{P}_k|} \left(\sum_{p\in S} \bar{g}_p + \mathcal{N}\big(0, \sigma^2 C^2 I\big) \right), \tag{5}$$

where the divisor is a public constant rather than the realized $|S|$, and the head is updated by $\theta_{\mathrm{head}} \leftarrow \theta_{\mathrm{head}} - \eta\, \tilde{g}$.

Each client uses the smallest noise multiplier that meets the training budget for its own sampling rate and step count,

$$\sigma_k = \min\big\{\sigma :\ \varepsilon_k\big(\sigma,\, q_k,\, T_k\big) \le \varepsilon\big\}, \tag{6}$$

where ε_k is obtained from the privacy loss distribution accountant [31], or from a Rényi accountant [19], [33] for the single-multiplier arms $\sigma^\star = \max_k \sigma_k$, and the step count is the number the client executes,

$$T_k = R\, \tau_e\, \max\!\Big(1, \Big\lfloor \frac{|\mathcal{P}_k|}{B} \Big\rfloor\Big) \tag{7}$$

for the patient sampler of Algorithm 1. T_k must equal the executed count; a count that differs from the executed one yields a budget that is violated without any observable symptom.

Aggregation applies the weighted average at each tier. Writing $\theta_k^{(r,s)}$ for the model of client k after edge round s of cloud round r,

$$\theta_e^{(r,s)} = \sum_{k:\, e(k)=e} \frac{n_k}{\sum_{j:e(j)=e} n_j}\, \theta_k^{(r,s)}, \tag{8}$$

$$\theta^{(r)} = \sum_{e=1}^{E} \frac{N_e}{\sum_{e'} N_{e'}}\, \theta_e^{(r,\tau_e)}, \qquad N_e = \sum_{k:e(k)=e} n_k. \tag{9}$$

Updates are encoded before transmission. Sparsification retains the κd largest-magnitude coordinates of an update Δ of dimension d,

$$[\Delta^{\mathrm{sp}}]_i = \begin{cases} [\Delta]_i, & i \in \arg\operatorname{top}_{\kappa d} |\Delta| \\ 0, & \text{otherwise}, \end{cases} \tag{10}$$

and the retained values are quantized uniformly to b bits with step $s = (v_{\max} - v_{\min})/(2^b - 1)$,

$$Q(v) = v_{\min} + \operatorname{round}\!\Big(\frac{v - v_{\min}}{s}\Big)\, s. \tag{11}$$

The residual $\Delta - \Delta^{\mathrm{sp}}$ is accumulated locally and added to the next round's update. Both operations act on the already-perturbed quantity of Eq. (5) and are therefore post-processing.

The guarantee is stated formally below. Two datasets are *neighboring* if one is obtained from the other by adding or removing every record of a single patient. Three conditions are assumed. (A1) Each patient's records lie in the training split of exactly one client, and the training, validation and test cohorts are disjoint in patients. (A2) The following are public: the architecture, the encoder parameters θ_{enc}, which are pretrained on a patient-disjoint public pool, all hyperparameters $(R, \tau_e, B, C, \sigma, \eta, \kappa, b)$, the edge map, the input standardization constants, and the declared client sizes $|\mathcal{P}_k|$ and n_k, which are fixed before training; q_k, T_k, σ_k and the aggregation weights are computed from them and never from the data. (A3) The forward pass of a record depends on that record alone; the model contains no normalization across records, such as batch normalization, so the gradient of one patient's records is unaffected by any other patient.

Theorem 1 (Patient-level guarantee): Under (A1)–(A3), let each σ_k satisfy Eq. (6). The transcript of a PP-HFM run, consisting of every encoded client update, every edge and cloud model, and the model selected on the validation cohort, is (ε, δ)-differentially private with respect to each training patient, under adding or removing all records of one patient with the parameters of (A2) held fixed.

Proof: Fix a training patient p, held by client k by (A1). *One step.* Given the model entering a step, the per-patient gradients $\bar{g}_{p'}$, $p' \neq p$, are unchanged by the presence of p by (A3), and $\|\bar{g}_p\|_2 \leq C$ by Eq. (4). The noisy sum in Eq. (5) is therefore a Poisson-subsampled Gaussian mechanism with rate q_k, noise σC and ℓ_2-sensitivity C under addition or removal of p, whose privacy loss is bounded in [31], [33]. Division by the public constant $q_k|\mathcal{P}_k|$ and the optimizer update are post-processing. *Composition at client k.* Each step takes as input the outputs of earlier steps, of other clients and public quantities, so the T_k steps compose adaptively, and Rényi divergences add [19]. The accountant evaluates the composed bound exactly for (σ, q_k, T_k), and Eq. (6) ensures $\varepsilon_k \leq \varepsilon$ at δ. Because σ_k is a function of the public $|\mathcal{P}_k|$ and T_k only, using a different multiplier at each client releases nothing. *Other clients.* By (A1) no other client accesses the records of p; its outputs are randomized functions of its own data, public quantities and messages already released, and therefore contribute no privacy loss with respect to p. *The hierarchy.* Sparsification with a locally stored residual, quantization, edge and cloud averaging with the public weights of Eqs. (8)–(9), and broadcast are functions of released outputs and public quantities, and selection on the validation cohort uses data disjoint from p by (A1); all are post-processing. ∎

Corollary 1 (Coincidence of the units): If every patient contributes one record, Algorithm 1 is differentially private stochastic gradient descent with Poisson sampling over records at rate q_k, with the same clipping norm and noise. For equal (B, T_k, σ, C) the patient-level and record-level mechanisms are the same mechanism, so a difference between them at one record per patient measures seed variation or an implementation difference, not a property of the privacy unit.

The theorem bounds what the guarantee reveals; the next statement predicts what it costs. In the scaling experiments of Section V-B3, P individuals each contribute exactly k records, so the record-level arm minimizes $F(\theta) = \frac{1}{Pk}\sum_p \sum_{w \in W_p} \ell_w(\theta)$ and the individual-level arm minimizes the same F written as $\frac{1}{P}\sum_p \bar{\ell}_p(\theta)$, with $\bar{\ell}_p = \frac{1}{k}\sum_{w \in W_p} \ell_w$.

Proposition 1 (Cost in records per individual): Let each ℓ_w be convex and L-Lipschitz on a parameter domain of diameter D in $\mathbb{R}^d$, and let both arms be trained by differentially private gradient descent at the same (ε, δ), and let $\Lambda = LD\sqrt{d\log(1/\delta)}/\varepsilon$. (i) The excess empirical risk of the record-level arm is at most $c_{\mathrm{rec}}\Lambda/(Pk)$ and that of the individual-level arm at most $c_{\mathrm{ind}}\Lambda/P$, with constants independent of k; both rates are tight in the worst case up to logarithmic factors. (ii) If both arms attain their rate with a common constant c, the gap is $A(1 - 1/k)$ with $A = c\Lambda/P$.

Proof: For n units with convex L-Lipschitz losses, private gradient descent attains excess empirical risk $O(LD\sqrt{d\log(1/\delta)}/(n\varepsilon))$, and this rate is tight up to logarithmic factors [34]. The record-level arm has $n = Pk$ units. In the individual-level arm the unit is $\bar{\ell}_p$, which is L-Lipschitz as an average of L-Lipschitz losses; clipping the summed gradient at C equals clipping the gradient of $\bar{\ell}_p$ at C/k, a rescaling that leaves the guarantee unchanged, so the mechanism is private gradient descent over $n = P$ units. This gives (i), and (ii) follows by subtraction. ∎

Part (ii) rests on an assumption that (i) does not supply: upper bounds with different constants do not determine the attained gap. The form $A(1 - 1/k)$ is therefore a *scaling hypothesis* that the proposition motivates, not a theorem, and it is tested empirically, including out of sample. Moreover, the proposition concerns bounds for convex losses, whereas the head is not convex and the reported metric is average precision rather than excess risk. The hypothesis therefore concerns the *shape* of the cost, not its magnitude: a gap of the form $A(1 - 1/k)$ with one free parameter, zero at $k = 1$, and an individual-level arm whose utility does not depend on k. It also identifies what would recover the cost. Additional records per individual reduce the variance of $\bar{\ell}_p$'s gradient, but a clipping bound fixed around zero cannot exploit this; mechanisms that clip around an estimate of the per-individual mean can, and attain user-level rates that improve with k [35]. Rescaling the summed gradient by the record count, tested in Section V-B6, is not such a mechanism.

The theorem fixes what is and is not covered. On eICU and on the person-level VitalDB partition the unit is the person. On the case-level VitalDB partition the unit is the surgical case: 230 of 5,974 people (3.9 %) underwent more than one operation, and for them the guarantee holds per operation and extends to the person only by group privacy over at most 14 cases, composed across clients for the 68 training persons whose cases lie on two. On eICU, where clients are hospitals, (A1) fails for the 2,177 of 74,327 training persons (2.9 %) whose stays lie at two to four hospitals. The guarantee applies directly to individuals confined to one hospital; for the others it is the composition of the mechanisms of every hospital that holds them, so the per-client budget alone does not establish a uniform $\varepsilon = 0.9$ for the whole eICU training population. These mechanisms compose adaptively; with the accountant of the headline, their composed training budget at $\delta = 10^{-5}$ is at most 1.32 for the 2,097 persons at two

TABLE III
SCOPE OF THE GUARANTEE IN EACH EVALUATED CONFIGURATION

Configuration	Unit	Encoders / standardization	Accounting	Guarantee covers
eICU, headline	patient	disjoint / training cohort	σ_k, PLD; $0.9 + 0.1$	training transcript‡
eICU, pool-fitted constants	patient	disjoint / public pool	σ_k, PLD; $0.9 + 0.1$	end to end‡
VitalDB, person-level headline	person	oracle / training cohort	σ_k, PLD; $0.9 + 0.1$	transcript; (A2) violated
VitalDB, disjoint encoders (case-level)	surgical case	disjoint / training cohort	σ_k, PLD; $0.9 + 0.1$	transcript, per case
VitalDB, cost of the unit and audit (person-level)	person vs record	oracle / training cohort	$\sigma_{\max}$, RDP; $1.0 + 0.5$	training transcript
Scaling grids, clipping rule (VitalDB case-level; eICU)	individual vs record	oracle (VitalDB), disjoint (eICU) / training	$\sigma_{\max}$, RDP; 0.8–$1.0 + 0.2$–0.5	training transcript
WEAR	record (= participant)	oracle / training cohort	record-level; $0.766 + 0.2$	transcript, per record

Unit: neighboring datasets differ by all records of one patient, person or case, or by one record; "vs" marks a comparison of two arms. Accounting: $\varepsilon_{\text{train}}+\varepsilon_{\text{conf}}$ at $\delta = 10^{-5}$; σ_k, per-client multipliers; $\sigma_{\max}$, one worst-client multiplier; PLD, privacy-loss-distribution accountant; RDP, Rényi accountant. The conformal release was evaluated and is not part of the deployed system. Arms compared with each other share one configuration. Other analyses (recipe, ablation, fusion, attribution) are labeled with their partition and mechanism in Section V. ‡For the $97.1\,\%$ of training persons at one hospital; the $2.9\,\%$ at two to four hospitals compose the mechanisms of those hospitals, $\varepsilon_{\text{train}} \leq 1.32$–$1.92$.

hospitals, 1.64 for the 75 at three and 1.92 for the 5 at four (Rényi accountant: 1.42, 1.77, 2.08). Likewise, the evaluated conformal release composes to at most $\varepsilon_{\text{conf}} = 0.2$ or 0.3 for the 509 validation persons ($2.6\,\%$) at two or three hospitals. On the wearable deployment each client is one participant, and the mechanism evaluated there is record-level; a person-level guarantee would require client-level privacy at the edge, which is not evaluated. Outside the guarantee are the validation cohort, which is used without noise for checkpoint selection, the input standardization constants, which are computed once from the training cohort, and the declared client sizes, which are treated as public, as is conventional for privately trained models [5]. Adding or removing a patient changes the realized cohort but not the mechanism, because every parameter is fixed by the declared sizes before training; this is the setting in which the Poisson-subsampled Gaussian is analyzed [31], [33]. In the experiments the declared sizes equal the realized cohort sizes; where a size is sensitive, it can be replaced by a coarse public value or released by a private count, and the theorem holds for whichever declared values are used. Table III lists, for each evaluated configuration, the unit, the provenance of the encoders and standardization constants, the accounting and what the guarantee covers. The implementation of the accounting is additionally checked by a suite of 17 regression checks, seven of which cover it: composition of the budget across calls, release and accounting of noise at every step with the expected-batch divisor, one conformal score per calibration patient, and per-client multipliers within the budget. These are implementation-level checks of the accounting configuration; they do not by themselves establish the guarantee.

Finally, the audit statistic of Section III-K is the logit-scaled confidence of a patient, averaged over that patient's records,

$$\varphi(p) = \frac{1}{|W_p|} \sum_{(x,y)\in W_p} \Big(z_y(x) - \log \sum_{j \neq y} e^{z_j(x)} \Big), \tag{12}$$

which is the numerically stable form of $\log\big(\hat{p}_y/(1-\hat{p}_y)\big)$, and membership is scored by the ratio of the fitted in- and out-densities,

$$\Lambda(p) = \log \frac{\mathcal{N}\big(\varphi(p);\, \mu_{\text{in}}(p), \sigma^2_{\text{in}}(p)\big)}{\mathcal{N}\big(\varphi(p);\, \mu_{\text{out}}(p), \sigma^2_{\text{out}}(p)\big)}. \tag{13}$$

The objective in Eq. (1) is minimized over the fusion head only, using the mechanism of Eqs. (3)–(5) with the noise scale given by Eq. (6), while the encoders remain frozen. Optimization uses stochastic gradient descent on the perturbed gradient. The reported model is selected by validation performance.

IV. EXPERIMENTAL SETUP

Fig. 5 outlines the study from partitioning to audit.

A. Datasets

The framework is evaluated on two publicly available multimodal corpora: VitalDB [24], a clinical monitoring collection of non-cardiac surgical cases, and WEAR [25], an outdoor activity collection combining body-worn inertial sensing with egocentric video. A third corpus, the eICU Collaborative Research Database [36], distributed through PhysioNet [37] under a credentialed data-use agreement, serves as an external clinical corpus. In all three corpora, the two modalities originate from the same individual and are paired at the source rather than by a construction rule. This property is necessary for a claim about fusion, since a fusion benefit under rule-based pairing may reflect the rule rather than complementary information. The corpora also cover the two deployment regimes relevant to the privacy mechanism: a client holding many individuals and a client holding one.

In VitalDB, modality A is the intraoperative monitoring record with eighteen channels, namely nine numeric physiological trends and nine binary indicators of whether each trend was recorded. Modality B is a thirty-four-dimensional vector of preoperative clinical and laboratory variables. All channels of a case originate from the same patient during the same operation and are time-aligned at capture across the waveform digitizer, the patient monitor and the anesthesia machine. The prediction target is postoperative admission to intensive care, which is positive in $18.84\,\%$ of windows ($19.65\,\%$ in the

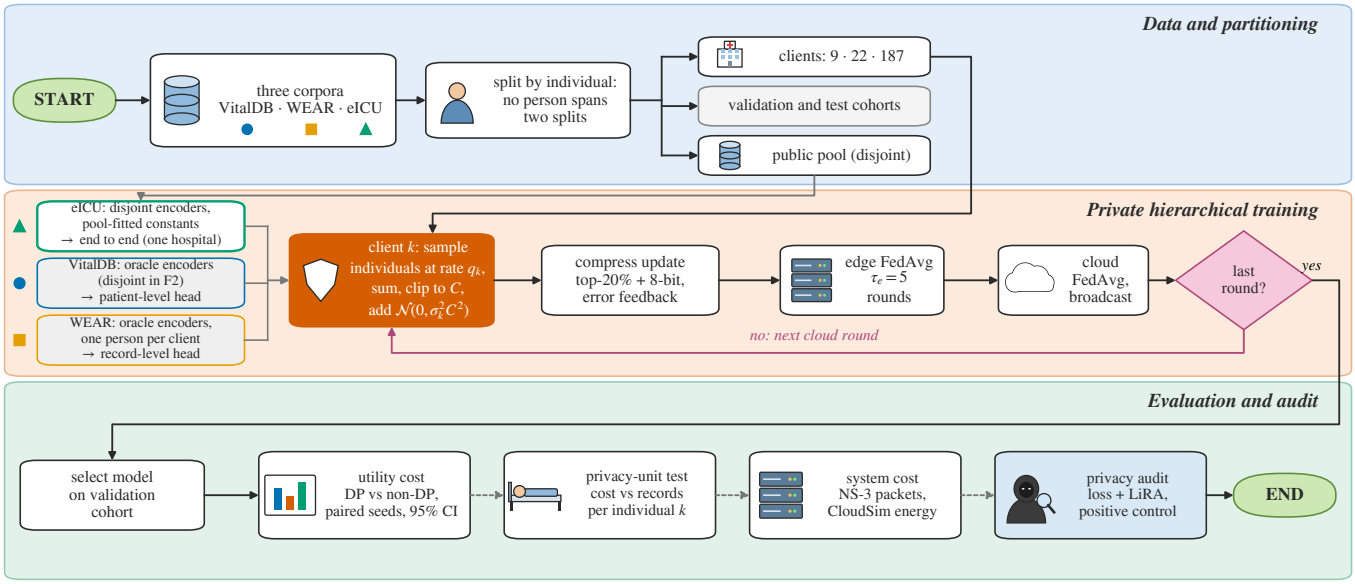


Fig. 5. Study workflow: individual-level partitioning, per-corpus encoder configuration (end to end on eICU only, for individuals at one hospital), private hierarchical training and paired evaluation with an attack audit.

case-level partition defined below). In WEAR, modality A consists of twelve inertial channels and modality B is a 2048-dimensional two-stream video feature computed over the same interval and synchronized in hardware. The task is activity recognition over eighteen classes.

In eICU, modality A consists of twelve channels: six vital signs recorded during the ICU stay (heart rate, respiratory rate, oxygen saturation, systolic and diastolic arterial pressure, and temperature) and six binary indicators of whether each was observed. The first 24 hours after unit admission are divided into non-overlapping 60-minute windows of twelve 5-minute slots, and a window is retained only if at least three slots are observed. Modality B is a five-dimensional vector recorded at or before admission (age, sex, admission weight, admission height and the APACHE score) for the same stay. The prediction target is in-unit mortality, with a prevalence of 0.0422. Individuals who die within the observation window are excluded, since their final windows are peri-mortem and pose no prediction problem. Prediction is therefore made at hour 24, when the APACHE score, computed from the worst values of the first day, is available. The resulting corpus contains 3,782,805 windows from 185,140 stays of 132,736 individuals.

Each corpus is partitioned into disjoint training, validation and test subsets, which are fixed across all experiments, so no record appears in more than one subset and all arms use the same split. VitalDB is evaluated on two partitions, both distributing cases across nine clients by surgical specialty and two edge aggregators. The *person-level* partition assigns each person to one split and one client; cases that would place a person in a second split or on a second client are removed (200 cases). It contains 33,491 training, 6,773 validation and 6,895 test windows from 4,221, 868 and 885 persons, with 7.9 training windows per individual on average. The *case-level* partition of the earlier experiments splits by operation (34,044, 7,308 and 7,360 windows from 6,257 operations); in it, 55 of 940 test persons also have a training case and 68 training persons span two clients, so its results protect the operation rather than the person. The privacy cost, the cost of the unit and the comparison with prior systems are person-level; all other VitalDB results are case-level and marked as such. No test case of either partition was used to pretrain the encoders. For WEAR, the partition contains 26,424 training, 5,395 validation and 5,969 test windows across twenty-two clients, one per participant, also assigned to two edges. Partitioning by specialty rather than at random produces realistic heterogeneity, since specialties differ in case mix, outcome prevalence and routinely recorded channels (Fig. 6). For eICU, the clients are the hospitals themselves, taken as given rather than constructed. Of 206 hospitals, those with fewer than 200 windows or with no positive outcome are pooled into a single client, giving 187 clients assigned to two edge aggregators, with a median of 444 individuals per client and 28.50 records per individual. Splits are made by individual, stratified by outcome within each client, because an individual may be admitted more than once, and a stay-level split would place the same person in both training and test data. Case-level metrics on eICU are grouped by stay rather than by individual, because the outcome of a person can differ between admissions. The validation subset is used only for model selection, and the test subset only for final evaluation.

Four controls are applied in preparing the clinical corpus, each preventing an inflation of the reported results that would not be visible during training. First, data are split by case, so all windows of an operation fall in the same subset; a window-level split would place highly correlated near-duplicates of training examples in the test set. Second, modality B is restricted to preoperative variables, and all intraoperative columns are excluded, because interventions such as transfusion or vasopressor administration are given in

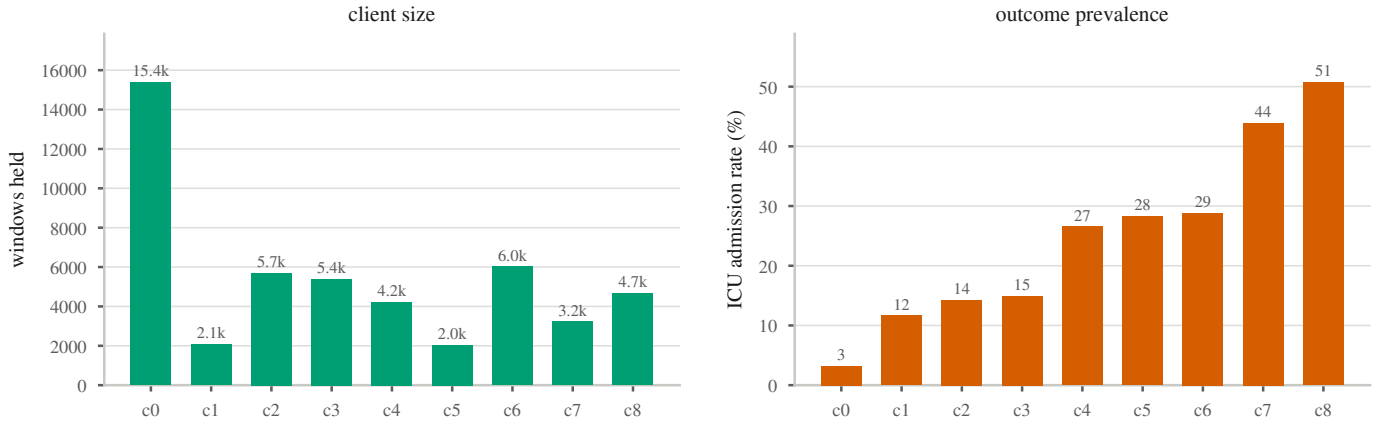


Fig. 6. Heterogeneity of the clinical partition (case-level). Partitioning by surgical specialty rather than at random yields clients that differ by a factor of seven in size and by more than fifteen in outcome prevalence, from $3.23\,\%$ to $50.72\,\%$. The skew is a property of how care is organized, not an artifact imposed for the experiment.

response to deterioration and therefore encode the outcome. Third, missing values are not simply imputed: gaps are filled within a case and paired with an explicit binary indicator, because, for example, an arterial line is placed only in patients judged to be at risk, so its absence carries information about acuity. Fourth, normalization statistics are computed over training cases only, with estimators that handle missing values, because a naive estimator returns an undefined statistic for a fully missing channel and propagates it to every window. In addition, a separate set of individuals, disjoint at the individual level from all client data, is reserved before federated training and used only to pretrain the encoders, so that the reported budget holds end to end. For eICU, this pool contains $526{,}867$ windows, leaving $2{,}122{,}350$ training, $563{,}948$ validation and $569{,}640$ test windows for the clients.

B. Implementation Details

All arms use the same training configuration, so that differences can be attributed to the mechanism under test. Each client performs one local pass per federated call, five edge rounds are executed per cloud round, and training runs for ten cloud rounds. The privatized gradient is passed to Adam (learning rate 10^{-3}, weight decay 10^{-4}), which is post-processing, with a per-individual clipping norm of one, 64 individuals sampled per step in expectation, and a noise multiplier per client from the calibration of (6). Updates are sparsified to the largest twenty percent of coordinates and quantized to eight bits before transmission, with error feedback of the residual. The full budget goes to training: $\varepsilon = 0.9$ on the clinical deployments. The encoders are pretrained centrally and then frozen, so the privatized parameters are those of the fusion head only: $559{,}874$ of $1{,}864{,}227$ parameters, or $30.0\,\%$.

Unless stated otherwise, all results, including the system-level objectives, the privacy–utility comparisons and the membership-inference audit, were obtained with the *oracle* encoder configuration of Section III-D, in which the encoders were pretrained on the pooled client training data. This configuration does not provide a valid end-to-end guarantee. It is used because it isolates the behavior of the privatized head from the origin of the representation, and because using the same encoders in every arm keeps the comparisons between arms controlled. The *disjoint* configuration, which provides the guarantee, is evaluated separately, and the difference is reported in Section V-A as the cost of a valid guarantee. On eICU, every experiment that uses pretrained encoders uses the disjoint configuration. Absolute utilities should therefore be read as upper bounds for the deployable configuration, whereas differences between arms, on which the claims rely, are unaffected. All normalization layers are group or layer normalizations, as explained in Section III-D.

Models are implemented in PyTorch 2.5, with per-sample gradients and Rényi accounting provided by Opacus 1.5. Deployment models are exported to ONNX at full precision, and inference is timed with ONNX Runtime on the CPU provider of a workstation (Intel Core i7-14700), since an edge node cannot be assumed to have an accelerator; no edge hardware was used. Network behavior is simulated in ns-3 3.46, with the device-to-edge segment over a wireless local-area link and the edge-to-cloud segment over a point-to-point backhaul. Energy is modeled in CloudSim Plus separately for the device, edge and cloud tiers. Training was performed on a single workstation with one NVIDIA RTX A2000 GPU. Table IV collects these parameters.

C. Evaluation Metrics

Classification performance is assessed with two complementary measures. The area under the receiver operating characteristic curve summarizes ranking performance across all decision thresholds. Average precision summarizes the precision–recall trade-off,

$$\mathrm{AP} = \sum_{n} \left(R_n - R_{n-1}\right) P_n, \tag{14}$$

where P_n and R_n are the precision and recall at the nth threshold. Average precision is reported because the clinical task is imbalanced, with $18.84\,\%$ positive windows. Under this imbalance, accuracy is nearly uninformative, since a constant negative prediction achieves about four-fifths accuracy, and

TABLE IV
TRAINING AND EVALUATION PARAMETERS

Parameter	Value
Cloud rounds R; edge rounds τ_e	10; 5
Edge aggregators E	2 (eICU scaling test: 2–16)
Local passes per edge round	1
Optimizer	Adam
Learning rate; weight decay	10^{-3}; 10^{-4}
Expected individuals per step B	64
Clipping norm C	1
Training budget	$\varepsilon = 0.9$, $\delta = 10^{-5}$
Accountant	privacy loss distribution, per-client σ_k
Records per patient (cap)	64
Update encoding	top-20 %, 8-bit, error feedback
Seeds	5 (paired); 3 for grids
LiRA shadow models	16
Latency timing	P95, 1,000 runs, 100 warm-up
Software	PyTorch 2.5, Opacus 1.5, ONNX Runtime, ns-3 3.46, CloudSim Plus 8.5
Hardware	one NVIDIA RTX A2000 GPU

the area under the ROC curve is optimistic. Average precision reflects performance on the minority class.

Metrics are also aggregated per individual. The model scores windows, but the clinical question concerns an operation, and windows of the same case are highly correlated, so window-level metrics overstate the effective sample size. The predictions of a case are therefore averaged,

$$\bar{p}_c = \frac{1}{|W_c|} \sum_{w \in W_c} p_w, \tag{15}$$

giving one prediction per individual. Case-level metrics are computed from $\bar{p}_c$ using the same expressions as for p_w.

Comparisons between arms are paired by seed, since the arms share their partition, initialization schedule and calibration data. For S paired seeds with per-seed difference d_s, the reported interval is

$$\bar{d} \pm t_{S-1,\,0.975} \frac{s_d}{\sqrt{S}}, \tag{16}$$

where s_d is the sample standard deviation of the differences. Student's t distribution is used instead of a normal approximation because the number of seeds is small. Seed counts are fixed before each comparison and are not increased after inspecting the results.

Communication, latency, reliability and energy are reported as defined in Section III-J, and every network quantity is measured on rounds in which the payload traversed the simulated topology.

D. Experimental Protocol

The proposed system is compared with differentially private federated averaging in three configurations, each isolating one design decision. The first trains the full parameter vector under the privacy budget, which is the conventional application of DP-SGD to a federated model. The second restricts private optimization to the fusion head over randomly initialized frozen encoders, isolating the effect of reducing the privatized dimension. The third restricts private optimization to the fusion head over *pretrained* frozen encoders. This arm controls for a confound: without it, the benefit of parameter-efficient private training cannot be separated from the benefit of the pretrained representation. A non-private model trained with an otherwise identical recipe provides the reference for the accuracy cost of the budget. All arms share the partition, schedule, compression settings and evaluation procedure, and differ only in the factor under test.

The same evaluation protocol is applied to every arm. The checkpoint with the highest validation performance is retained, without use of test data. Each retained checkpoint is evaluated once on the test subset, and all metrics are computed from the same predictions with the same aggregation.

The privacy guarantee is audited by individual-level membership inference using the two attacks of Section III-K. Each attack is also applied to a configuration known to memorize, namely non-private training of the same head over the same frozen encoders, as a positive control. A negative result is accepted only if the same procedure detects clearly elevated leakage on the control.

A further experiment separates the two sources of the cost of the individual-level unit. Moving the guarantee from the record to the individual both reduces the number of units sampled per step and bounds the summed contribution of each individual, and the aggregate comparison cannot separate the two. The number of units sampled per step is therefore held equal in both arms, which removes the first effect and leaves only the clipping geometry, and the number of training records retained per patient, k, is varied over $\{1, 2, 4, 8\}$ with three seeds at $\varepsilon = 1$. Subsampling is applied to the training partition only; validation and test sets are unchanged, so every configuration is evaluated on the same held-out data. Because reducing k also reduces the training set size, absolute utility decreases with k and is not comparable across k. The quantity of interest is the paired difference between arms at each k, where both arms use the same data, partition and seed. At $k = 1$, a patient contributes a single record, so the two arms must agree, which provides a falsification test of the design.

V. RESULTS

A. System-Level Objectives

The framework is evaluated on three deployments that differ in client composition. In the clinical deployment, nine institutional clients hold the records of $5{,}974$ persons, so each client contains many individuals, and each individual contributes 7.9 training records on average. In the wearable deployment, each of the twenty-two clients corresponds to a single participant who contributes approximately $1{,}750$ records. The second deployment also tests whether the system-level behavior generalizes across topology and payload size. In the external deployment, 187 hospital clients share two edge aggregators, which places about twenty times as many stations on each wireless medium as in the clinical deployment.

Table V reports the measured value of each design objective of Table I. All targets are met on the clinical deployment. On the wearable deployment, all targets are met except the

TABLE V

OUTCOMES, MEASURED OR SIMULATED, AGAINST THE PRE-REGISTERED TARGETS OF TABLE I. UPLINK PAYLOAD IS MEASURED FROM THE DEPLOYED COMPRESSOR; DELIVERY AND ROUND LATENCY ARE SIMULATED AT PACKET LEVEL IN NS-3 ON EVERY ROUND; ENERGY IS MODELED IN CLOUDSIM PLUS; INFERENCE LATENCY IS TIMED ON A WORKSTATION CPU. O2A IS THE TRAINING BUDGET; THE EVALUATED, NON-DEPLOYED CONFORMAL RELEASE ADDS $\varepsilon_{\text{conf}} = 0.1$ ON THE CLINICAL DEPLOYMENTS (TOTAL 1.0) AND 0.2 ON THE WEARABLE DEPLOYMENT (TOTAL 0.966). O2B IS A PAIRED COMPARISON AGAINST A NON-PRIVATE REFERENCE TRAINED UNDER THE IDENTICAL RECIPE, FIVE SEEDS, IN POINTS OF AREA UNDER THE ROC CURVE ON THE TWO CLINICAL DEPLOYMENTS (THE PRIVATE MODEL PREDICTS THE MAJORITY CLASS AT THE DEFAULT THRESHOLD, SO ACCURACY CANNOT MEASURE THE LOSS); §ON THE WEARABLE DEPLOYMENT, WHOSE GUARANTEE IS RECORD-LEVEL, IT IS THE LOSS IN ACCURACY OVER FIFTEEN SEEDS. †AN INTERVAL THAT SPANS ITS THRESHOLD, INCONCLUSIVE AT THE PRE-REGISTERED SAMPLE SIZE (EXTERNAL DEPLOYMENT: $[1.37, 4.81]$, POINT ESTIMATE ABOVE THE THRESHOLD; END TO END 3.91 $[2.40, 5.42]$, NOT MET). *NOT MET. CLIN., WEAR., EXT.: CLINICAL, WEARABLE AND EXTERNAL DEPLOYMENTS. CLINICAL O2A AND O2B ARE PERSON-LEVEL, OTHER ROWS CASE-LEVEL.

	Quantity	Clin.	Wear.	Ext.	Target
O1	Uplink reduction	90.8%	96.1%	92.3%	$\geq 50\%$
O1b	Round latency	16.9%	21.7%	33.4%	$> 0\%$
O2a	Budget ε	0.900	$0.766^{\S}$	0.900	≤ 1
O2b	Utility loss [pts]	-1.18	1.97 pp†§	$3.09^{\dagger}$	≤ 2
O3a	P95 latency [ms]	2.30	0.75	0.24	< 100
O3b	Update success	100.00%	99.94%	97.83%*	$> 99\%$
O4	Energy saving	72.3%	76.8%	79.6%	$\geq 20\%$

TABLE VI

INFERENCE LATENCY DECOMPOSITION ON THE CLINICAL DEPLOYMENT, TIMED ON A WORKSTATION CPU (INTEL CORE I7-14700, NOT EDGE HARDWARE), 1000 TIMED EXECUTIONS AFTER 100 DISCARDED WARM-UP EXECUTIONS. THE LAST ROW REPORTS THE EIGHT-BIT QUANTIZED GRAPH, WHICH WAS EVALUATED AND NOT DEPLOYED.

Stage	Mean (ms)	P95 (ms)
Preprocessing	0.106	0.110
Forward pass	2.130	2.165
Conformal set (not deployed)	0.019	0.021
Post-processing	0.007	0.008
Total	2.263	2.299
Total, eight-bit graph	3.515	17.337

accuracy bound, which is inconclusive. On the external deployment, five of seven targets are met; the utility bound is inconclusive in area under the ROC curve under the prespecified interval criterion, with a point estimate above it, is met in balanced accuracy, and is missed end to end; update delivery falls short of its target.

All network quantities are measured on every simulated round. An earlier simulator configuration evaluated one round in ten at packet level and composed the remaining rounds analytically. The two approaches disagree: the analytical model placed the communication term of the centralized baseline at 256 ms, whereas the simulated network delivers 23.6 ms, which overstated the latency benefit of the hierarchy by a factor of about four. No analytically composed rounds contribute to the values reported here.

On both clinical deployments the training budget is $\varepsilon = 0.900$ under the privacy loss distribution accountant, against a requirement of one. Each client trains with its own noise multiplier (Section III-F); on the clinical deployment these range from 6.34 to 15.56, and on the external deployment from 4.79 to 29.34. On the wearable deployment, whose guarantee is record-level, it is 0.766.

A private calibration threshold. A conformal threshold released per client by the exponential mechanism over ranks [28], [32] ($\varepsilon_{\text{conf}} = 0.1$, $\alpha = 0.1$) was evaluated and is not part of the system. With about a hundred calibration patients per client, the rank utility barely separates candidate thresholds; the released threshold lies mid-range (0.42–0.61) while the private model's positive-class probabilities lie near zero (99th percentile 0.025 on the clinical deployment), so the prediction set is the majority class on every clinical seed and on four of five external seeds, and coverage equals the base rate (80.1 % on the clinical deployment, case-level partition; 95.7 % on the external one), missing the 90 % target. The non-private threshold does not collapse to the majority class (positive-class coverage 36–43 %), but its coverage, 80.4–81.8 %, also misses the target, so the shortfall lies mainly in the release rather than in the training noise; pooling calibration across clients is the expected remedy.

1) Communication and latency: Two-tier aggregation reduces uplink traffic by 90.8 % relative to centralized training, against a target of 50 %. The measurement is taken from the deployed compressor and includes the index array that top-κ sparsification transmits with the retained values, which analytical payload estimates omit and which is not negligible at high sparsity. The reduction is 96.1 % on the wearable and 92.3 % on the external deployment.

The hierarchy also reduces the end-to-end latency of a training round from 90.5 ms to 75.2 ms in the clinical deployment, a reduction of 16.9 %. The reduction is positive in every simulated round, ranging from 8.2 % to 25.8 %. It is smaller than the traffic reduction because the communication term falls from 23.6 ms to 16.5 ms, while the compute terms (sensing, preprocessing, inference and aggregation) are common to all topologies and dominate the total. The hierarchy reduces link time, not computation.

In the wearable deployment, round latency falls from 104.5 ms to 81.8 ms, or 21.7 %, with the communication term falling from 37.6 ms to 23.2 ms. The larger reduction is consistent with the larger topology: twenty-two stations contend for the same wireless medium rather than nine, so uplink latency is a larger share of the total. The external deployment continues the trend: with 187 clients, round latency falls from 261.4 ms to 174.1 ms, or 33.4 %, with the communication term falling from 194.6 ms to 115.5 ms, and the reduction is positive in every simulated round, ranging from 24.6 % to 40.9 %. The latency benefit of the hierarchy therefore grows with the number of stations that share a medium.

2) Cost of disjoint encoders: The results above were obtained with the oracle encoder configuration, in which the frozen representation was pretrained on the pooled client training data. This configuration does not satisfy assumption (A2), and its utility is therefore an upper bound. With the disjoint configuration, in which the encoders are pretrained only on a population reserved before federated training and

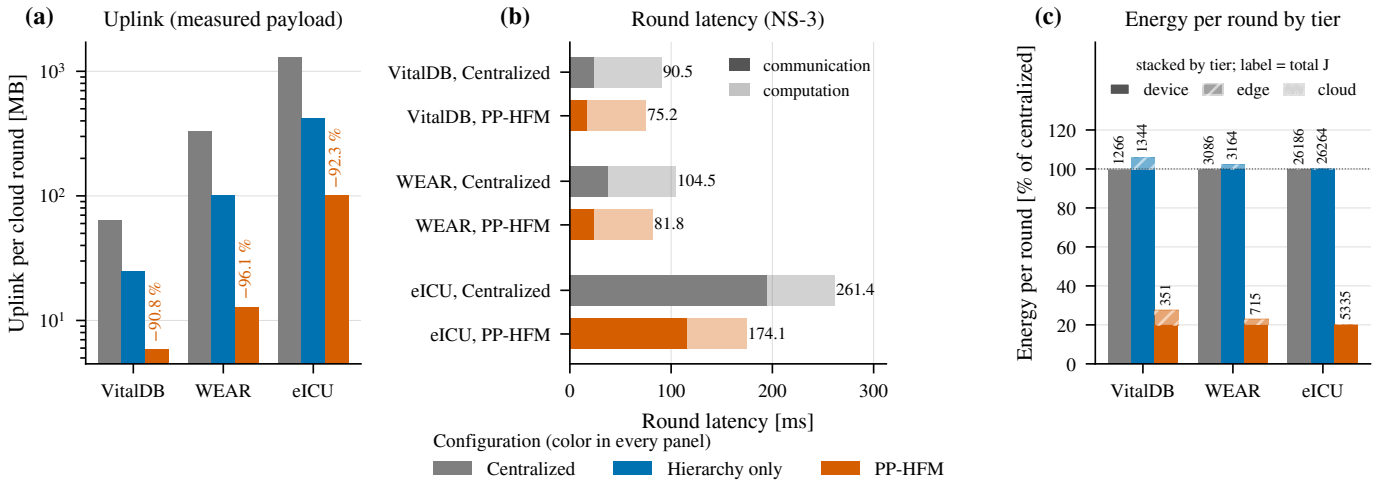


Fig. 7. System cost per deployment. (a) Uplink per cloud round, measured from the deployed compressor, for centralized training, the hierarchy with the full model, and the proposed configuration. (b) End-to-end round latency from packet-level simulation, split into communication (solid) and computation (light); the hierarchy shortens link time while computation is unchanged. (c) Energy per round by tier (device, edge, cloud) as a share of centralized training, with absolute values in joules. The hierarchy alone adds edge energy without reducing device energy; only the proposed configuration reduces the energy of the constrained tier.

disjoint from every client, the patient-level model (case-level partition) loses 0.0157 ± 0.0046 in area under the ROC curve ($0.8395 \rightarrow 0.8237$), paired over five seeds ($p < 0.001$), and 0.0122 ± 0.0262 in average precision, which is not resolved. Against a non-private reference trained with the same disjoint encoders and recipe, the cost of privacy is $+0.0014$ $[-0.0033, +0.0062]$ in area under the ROC curve and $+0.0221$ $[-0.0003, +0.0446]$ in average precision, at a training budget of $\varepsilon = 0.9$. This configuration satisfies (A2); its standardization constants are still fitted on the training individuals, so it is not end to end, which only the external-corpus rerun with pool-fitted constants is. Comparisons between arms are unaffected, because all arms share the same encoders.

3) Privacy and utility: The training budget of 0.900 satisfies the requirement $\varepsilon \leq 1$. The cost of the guarantee is measured against a non-private reference trained with the identical recipe, paired over five seeds. The measure was fixed before the runs: area under the ROC curve is primary, with average precision and balanced accuracy, at a threshold chosen on the validation cohort by Youden's index, reported alongside. Accuracy at the default threshold cannot serve: on both clinical deployments the private model predicts the majority class at 0.5 on every seed, so its accuracy equals the base rate (0.8113 and 0.958) and measures the threshold rather than the model.

On the clinical deployment no cost is incurred. The private model attains 0.8318 in area under the ROC curve against 0.8200 for the reference, a change of -1.18 points $[-1.63, -0.73]$ ($p = 0.002$); average precision changes by -3.47 points $[-6.33, -0.61]$ and balanced accuracy by -0.31 points $[-1.23, +0.62]$, negative values favoring the private model. The objective is met. The gain is consistent with regularization of the compact head by clipping and noise, which the case-level ablation suggests (Section V-D) but this experiment does not isolate; on the case-level partition the change was -0.90 points $[-1.82, +0.02]$.

On the external deployment a cost is resolved. The area under the ROC curve falls from 0.8245 to 0.7936, by 3.09 points $[1.37, 4.81]$. The interval spans the two-point target, so under the prespecified interval criterion the objective is inconclusive at five seeds, although the point estimate exceeds the target. Average precision falls by 1.72 points $[0.84, 2.60]$, an interval that spans the threshold, and balanced accuracy by 0.50 points $[-0.18, +1.18]$, which meets it. The cost is set by the smallest hospitals: a client of fifteen patients samples every patient at every step and requires the largest noise multiplier, 29.34, against a median of 13.39, and its updates enter the edge average with their public weights. With standardization constants fitted on the disjoint pool and encoders pretrained in those units, the only configuration in which the guarantee holds end to end for individuals at a single hospital, the cost rises to 3.91 points $[2.40, 5.42]$ (0.8209 to 0.7819, $p = 0.002$), with 2.66 $[1.82, 3.49]$ in average precision and 2.69 $[1.57, 3.82]$ in balanced accuracy. The end-to-end guarantee therefore misses the objective in area under the ROC curve, whose interval lies above the target; in average precision and balanced accuracy the intervals span it, with point estimates above it.

On the wearable deployment, where the guarantee is record-level, the loss in accuracy is 1.967 pp with an interval of $[1.760, 2.173]$ over fifteen seeds. The interval spans the 2 pp threshold, so this objective is *inconclusive*. The seed count was fixed in advance and was not increased after inspecting the interval.

4) Quality of service: Table VI decomposes the inference latency. The threshold step of the evaluated conformal set, which is not part of the deployed system, takes 0.019 ms of the 2.263 ms mean, or $0.84\,\%$, and is kept in the total as a conservative upper bound. The differentially private mechanism adds no inference cost, because its noise is applied to gradients during training and changes only the parameter values.

Timed on a workstation CPU rather than edge hardware, the

95th-percentile latency is 2.30 ms in the clinical deployment, 0.75 ms in the wearable deployment and 0.24 ms in the external deployment, against a budget of 100 ms, all measured on the full-precision graph on a CPU provider. This margin of more than fortyfold is due to the compact fusion head, not to the hierarchy or the privacy mechanism. Eight-bit dynamic quantization does not improve it: on the clinical deployment it raises the mean latency from 2.263 to 3.515 ms and the 95th percentile from 2.30 to 17.34 ms, and on the wearable deployment the quantized recurrent encoder does not yield a valid graph. The deployed model is therefore the full-precision graph. The difference between the deployments follows from the input dimension. The margin allows a larger backbone to be used within the same latency budget.

Update delivery succeeds for 100.00 % of transmissions in the clinical deployment and 99.94 % in the wearable deployment, against a target of 99 %. Both values are measured on every round. Under the earlier configuration, in which nine rounds in ten were composed analytically, the clinical value was 99.27 %. The wearable deployment does not reach perfect delivery, with a per-round minimum of 99.15 %, as expected when twenty-two stations share one medium.

The external deployment does not meet the target: 97.83 % of updates are delivered, with a per-round minimum of 96.81 %. This is a property of the topology rather than of the protocol. With 187 clients on two edges, every configuration loses packets to contention, and centralized training delivers only 95.11 %. The hierarchy improves delivery by 2.7 points but does not close the gap. Adding edge aggregators does. With the same 187 clients and payload, and only the edge count varied, delivery by the proposed configuration rises monotonically from 98.42 % at three edges to 99.94 % at sixteen. The mean meets the target from six edges, about 31 clients per edge, and every round meets it from eight edges, about 23 clients per edge, with a per-round minimum of 99.49 %. Round latency falls with contention, from 174.2 ms at two edges to 97.1 ms at eight. At sixteen edges, centralized training delivers 99.60 % against 99.94 % for the proposed configuration, so the advantage of the hierarchy persists once contention no longer limits delivery.

5) Energy: Energy per round decreases from 1266.2 J to 350.8 J in the clinical deployment, a saving of 72.3 %, and from 3086.2 J to 714.8 J in the wearable deployment, a saving of 76.8 %, and from 26,186.2 J to 5,334.8 J in the external deployment, a saving of 79.6 %, against a target of 20 %. Energy is reported per tier because a hierarchy redistributes computation rather than eliminating it (Fig. 7). Introducing the edge tier *alone* leaves device energy unchanged and adds edge energy, so the total increases from 1266 J to 1344 J. The saving occurs only in the configuration that also reduces device-side computation and transmission, which lowers the device term fivefold.

B. The Privacy Unit and the Budget

Two comparisons quantify the cost of the design decisions of Section III-E. Both are paired by seed against an arm that differs only in the factor under test.

TABLE VII
DECOMPOSITION OF THE RECIPE IN MACRO AUROC, THREE SEEDS (CLINICAL DEPLOYMENT: CASE-LEVEL PARTITION). THE INITIALIZATION STEP B4→B7 HOLDS THE PRIVATIZED SUBSET AT THE WHOLE NETWORK AND CHANGES ONLY THE STARTING POINT; THE SUBSET STEP B7→B6 HOLDS THE STARTING POINT AND PRIVATIZES 30 % OF THE NETWORK INSTEAD OF ALL OF IT, AND IS REPORTED AS MEAN ± A STUDENT-t HALF-WIDTH BECAUSE IT IS THE STEP THE CLAIM TURNS ON. THE CLINICAL ARMS ARE TRAINED UNDER THE PATIENT-LEVEL GUARANTEE; THE WEARABLE ARMS ARE RECORD-LEVEL (ONE PARTICIPANT PER CLIENT).

	Clinical		Wearable	
ε	B4→B7	B7→B6	B4→B7	B7→B6
∞	$+0.0129$	$+0.0139 \pm 0.0124$	$+0.0021$	-0.0002 ± 0.0003
8	$+0.1407$	-0.0025 ± 0.0059	$+0.0883$	$+0.0001 \pm 0.0002$
4	$+0.1569$	-0.0029 ± 0.0063	$+0.1257$	$+0.0001 \pm 0.0002$
2	$+0.1727$	-0.0035 ± 0.0040	$+0.1918$	$+0.0002 \pm 0.0001$
1	$+0.1841$	-0.0059 ± 0.0063	$+0.2827$	$+0.0003 \pm 0.0008$
0.5	$+0.1991$	-0.0042 ± 0.0043	$+0.3580$	$+0.0013 \pm 0.0088$

1) Cost of moving the unit from the record to the individual: Enforcing the guarantee over individuals rather than records changes the area under the ROC curve by -0.0170, with a paired 95 % interval of $[-0.0269, -0.0071]$ over five seeds ($0.8423 \rightarrow 0.8254$), and average precision by -0.0582 $[-0.0701, -0.0462]$; both intervals exclude zero. On the case-level partition the changes were -0.0147 and -0.0783. The cost is larger in average precision than in the area under the ROC curve, consistent with the minority class being harder to fit under a coarser sampling unit.

Two effects contribute to this cost. Sampling patients rather than records reduces the number of sampled units from 33,491 records to 4,221 patients on this partition, and clipping a patient's summed gradient constrains individuals with many records more strongly than those with few. Section V-B3 isolates the second effect by holding the number of units per step equal in both arms.

2) Initialization, not the privatized subset: This comparison and the scaling grids that follow are case-level. The proposed recipe differs from conventional private training in two respects: it starts from pretrained encoders, and it privatizes only the fusion head. A two-arm comparison cannot separate these factors, so three arms are compared. B4 is full-model private training from a random initialization, the prior-art configuration. B7 is full-model private training from the pretrained initialization, which changes only the initialization. B6 is head-only private training from the pretrained initialization, which changes only the privatized subset. Fig. 8 shows both steps at every budget on both deployments, and Table VII reports them over three seeds.

Almost the entire effect is due to the initialization, and it increases as the budget tightens. On the clinical corpus, the change from random to pretrained initialization is worth 0.0129 without noise and 0.1991 at the tightest budget, increasing monotonically in between; on the wearable corpus it ranges from 0.0021 to 0.3580. Under privacy, restricting the privatized subset changes macro AUROC by between -0.0059 and -0.0025 on the clinical corpus and by at most 0.0013 on the wearable corpus, and its intervals cover zero at every

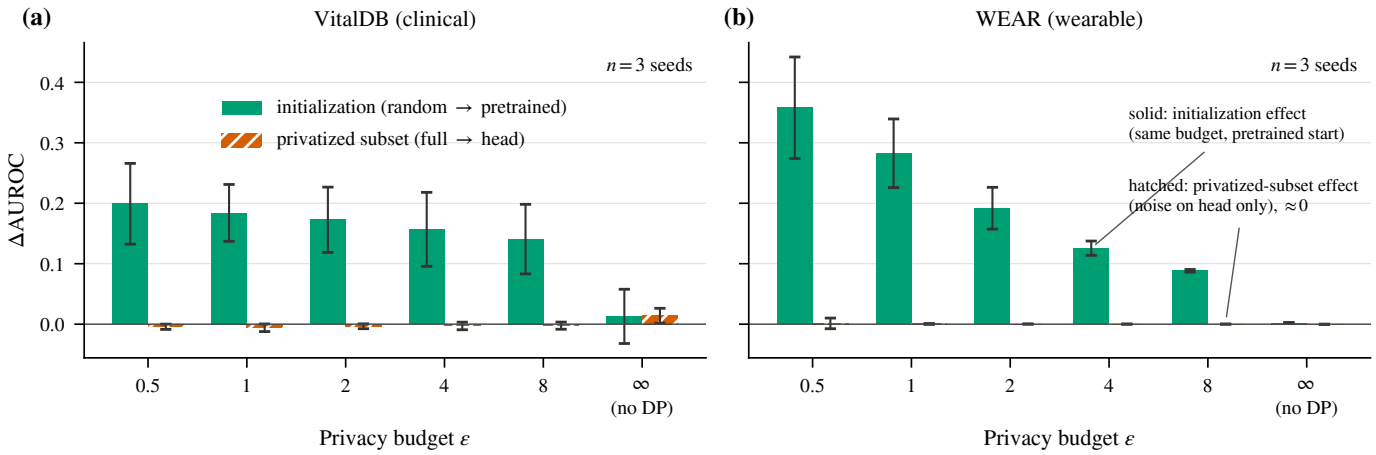


Fig. 8. Decomposition of the recipe's advantage across the privacy budget (area under the ROC curve, $n = 3$ seeds, Student-t 95 % interval; VitalDB case-level). Solid: changing only the initialization of the full model, from random to pretrained. Hatched: changing only the privatized subset, from the full pretrained model to the head. The subset step is indistinguishable from zero at every budget, although it privatizes 3.3 times fewer parameters.

finite budget. Only without noise does the head-only arm score higher on the clinical corpus, by 0.0139 ± 0.0124; no privacy mechanism is involved there, and the difference reflects fine-tuning the encoders on the clients.

This result contradicts a dimensionality explanation. If the advantage arose because a smaller parameter vector absorbs less noise, B7, which privatizes all 1,864,227 parameters, should perform worse than B6, which privatizes 559,874, and the difference should be largest at the tightest budget. At $\varepsilon = 0.5$, however, the two arms differ by only 0.0042, with an interval covering zero and with B7 the higher of the two, although B7 privatizes 3.3 times as many parameters under the same budget. This is consistent with the noise scale being determined by the budget, the number of steps and the sampling rate rather than by the number of parameters.

The same pattern appears in sensitivity to the budget. Across the full range, the randomly initialized arm loses 0.1906 on the clinical corpus and 0.3716 on the wearable corpus, where it falls to 0.6247, close to chance. The pretrained arms lose 0.0043 (full model) and 0.0225 (head only) on the clinical corpus and less than 0.016 on the wearable corpus. Robustness to a tight budget is therefore determined by the initialization. Under a fixed budget, a legitimate pretrained initialization should take priority over the choice of privatized parameters.

Restricting the privatized subset does reduce communication. Only head parameters are transmitted, which produces the uplink reductions reported in Section V-A, and Table VII shows that this comes at no measurable cost in accuracy. It is not, however, an accuracy mechanism, and a two-arm comparison of head-only training from a pretrained initialization against full-model training from a random initialization does not support such a claim, because the two arms differ in both respects.

3) Records per individual: Because a patient-level guarantee bounds the *summed* contribution of an individual, its cost is expected to increase with the number of records per individual and to vanish when each individual contributes a single record. Table VIII reports both arms as k, the number of training records retained per patient, is varied over $\{1, 2, 4, 8\}$ with three seeds. The number of units sampled per step is equal in both arms, which removes the sampling-rate effect, so only the clipping geometry differs. The validation and test partitions are unchanged.

TABLE VIII
AVERAGE PRECISION BY RECORDS PER INDIVIDUAL k (VITALDB CASE-LEVEL). THE GAP IS PAIRED BY SEED AND REPORTED AS MEAN $\pm$ THE HALF-WIDTH OF A STUDENT-t INTERVAL OVER THREE SEEDS; AN INTERVAL EXCLUDING ZERO IS ONE WHOSE HALF-WIDTH IS SMALLER THAN ITS MEAN.

k	Record	Individual	Gap	p
1	0.5015	0.5086	-0.0071 ± 0.0599	0.661
2	0.5590	0.5104	$+0.0486 \pm 0.0129$	0.004
4	0.5924	0.5128	$+0.0796 \pm 0.0163$	0.002
8	0.5955	0.5083	$+0.0872 \pm 0.0248$	0.004

At $k = 1$, the two arms are indistinguishable, with a gap of -0.0071 and an interval covering zero, as required by the construction. The divergence at larger k can therefore be attributed to the privacy unit rather than to the experimental setup. From $k = 2$ onward, the gap is resolved and increases monotonically, reaching 0.0872 at $k = 8$; the increment from $k = 4$ to $k = 8$ is an order of magnitude smaller than the preceding one (Fig. 9(a)).

The individual-level arm remains flat in k, with means within $[0.5083, 0.5128]$, whereas the record-level arm increases from 0.5015 to 0.5955. The widening gap therefore reflects a lack of improvement under the stricter unit rather than a degradation (Fig. 10).

Additional records per patient provide signal that the record-level arm exploits and the individual-level arm cannot, because the bound on the summed contribution is tightest for the individuals who contribute the additional records. On this corpus, collecting more data per patient therefore increases the relative cost of an individual-level guarantee.

The area under the ROC curve changes in the same direction, by 0.0026, 0.0085, 0.0159 and 0.0137 over the same values of k, but every interval covers zero at three seeds. The

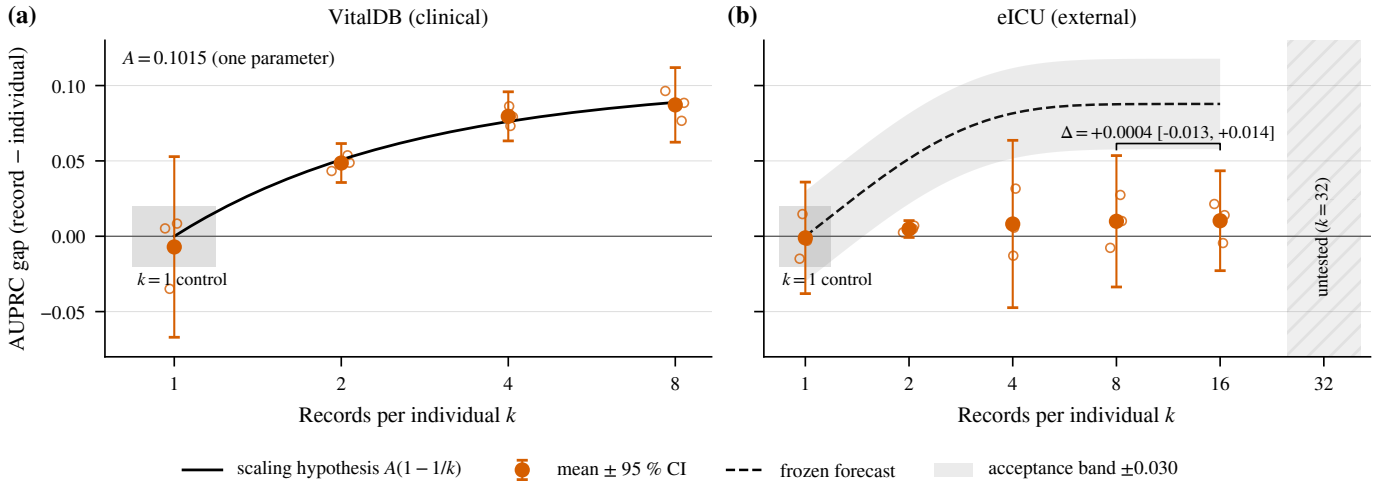


Fig. 9. Cost of the individual-level guarantee against the number of records per individual k, as the average precision of the record-level arm minus that of the individual-level arm, paired by seed. Open circles: seeds; filled: mean ± Student-t 95 % interval ($n = 3$). The grey box at $k = 1$ is the ±0.020 control tolerance. (a) VitalDB (case-level) with the fitted curve $A(1 - 1/k)$. (b) eICU against the law $A(1 - e^{-b(k-1)})$, frozen before the corpus was built, with its ±0.030 acceptance band; the bracket is the paired change from $k = 8$ to $k = 16$.

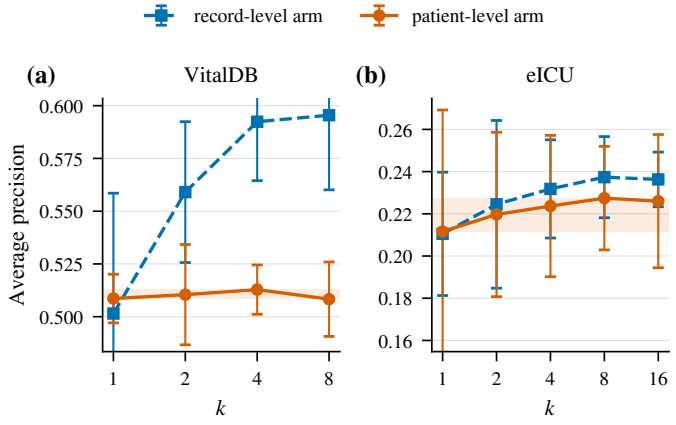


Fig. 10. Average precision of each arm against k, mean ± Student-t 95 % interval ($n = 3$; VitalDB case-level). The shaded band is the range of the individual-level means. The individual-level arm is flat on both corpora; the widening gap of Fig. 9 is the record-level arm improving. Both panels span 0.12 so that slopes are comparable.

effect is therefore resolved in average precision only. Accuracy is not reported for these arms, because an uncalibrated decision threshold on an imbalanced corpus keeps it near the majority rate; an area under the ROC curve of approximately 0.835 confirms that the models rank well.

Test of Proposition 1: The proposition predicts a gap of the form $A(1 - 1/k)$, with a single free parameter. Fitted to the twelve seed-paired gaps of Table VIII, it gives $A = 0.1015$ at a root-mean-square error of 0.0119, against 0.0117 for the two-parameter saturating law of Eq. (1): the extra parameter buys nothing. Fitted on $k \leq 4$ only, it predicts a gap of 0.0904 at $k = 8$, against 0.0872 observed. The individual-level arm is flat, as the proposition requires, with a slope of less than 0.001 per doubling of k. At the operating point of the external test ($\varepsilon_{\text{train}} = 0.8$, batch 64; Section V-B5) the same out-of-sample prediction is 0.0988 against 0.0914 observed, although that grid fails its $k = 1$ control.

4) External validation of the records-per-individual law: The law of Section V-B3 was measured on a single corpus whose clients are surgical specialties rather than institutions, so it is not established whether it characterizes patient-level private federated learning in general or only that cohort. It is therefore tested on a second clinical corpus of different construction: 3,782,805 hourly windows from 132,736 individuals across 187 clients that are *real hospitals*, with a median of 444 individuals per client and 28.50 records per individual. The label is in-unit mortality, with a prevalence of 0.0422.

Two properties of this replication are relevant. First, its encoders were pretrained on 526,867 windows verified to be disjoint from every client's training data, so the results do not carry the oracle-configuration caveat that applies elsewhere in this paper. Second, the prediction was fixed before the corpus was constructed. Fitting the four VitalDB points of Table VIII gives

$$\text{gap}(k) = A\left(1 - e^{-b(k-1)}\right), \tag{1}$$

with $A = 0.0878 \pm 0.0035$ and $b = 0.883 \pm 0.129$ at a root-mean-square error of 0.0028. This curve, together with an acceptance band of ±0.030 derived from the VitalDB seed spread, was registered before any record of the second corpus was accessed.

The control holds. At $k = 1$, the gap is −0.0011, with an interval covering zero and a magnitude within the pre-specified tolerance of 0.020. The two arms are therefore not separated where the construction requires them to agree, and the remaining rows can be attributed to the privacy unit.

The point forecasts are not confirmed. At every $k > 1$, the gap lies outside the ±0.030 band, and the measured cost is approximately one ninth of the predicted cost: 0.0099 against 0.0876 at $k = 8$.

Two properties of the law are reproduced. The first is the ordering: the gap is negative at $k = 1$, positive at $k = 2$, and

TABLE IX
AVERAGE PRECISION BY RECORDS PER INDIVIDUAL k ON THE EXTERNAL CORPUS, THREE SEEDS, PAIRED BY SEED. INTERVALS ARE STUDENT-t HALF-WIDTHS AS ELSEWHERE. THE ROW $k = 16$ LIES BEYOND THE RANGE OF THE FIRST CORPUS. THE FINAL COLUMN IS THE VALUE PREDICTED BY EQ. (1), FIXED BEFORE THE CORPUS WAS BUILT.

k	Record	Individual	Gap	p	Predicted
1	0.2105	0.2116	-0.0011 ± 0.0369	0.909	0.0000
2	0.2245	0.2197	$+0.0048 \pm 0.0057$	0.068	0.0515
4	0.2318	0.2237	$+0.0081 \pm 0.0554$	0.593	0.0816
8	0.2374	0.2274	$+0.0099 \pm 0.0437$	0.431	0.0876
16	0.2363	0.2260	$+0.0103 \pm 0.0332$	0.314	0.0878

TABLE X
AVERAGE PRECISION GAP (RECORD ARM MINUS INDIVIDUAL ARM, PAIRED BY SEED, THREE SEEDS) ACROSS THE THREE GRIDS. THE FIRST TWO DIFFER ONLY IN THE TRAINING BUDGET. THE FORECAST COLUMN IS EQ. (1), FITTED ON THE FIRST COLUMN AND NEVER REFITTED.

k	Forecast	First corpus $\varepsilon_{\mathrm{tr}} = 1.00$	First corpus $\varepsilon_{\mathrm{tr}} = 0.80$	Second corpus $\varepsilon_{\mathrm{tr}} = 0.80$
1	0.0000	-0.0071	$+0.0398$	-0.0011
2	0.0515	$+0.0486$	$+0.0364$	$+0.0048$
4	0.0816	$+0.0796$	$+0.0981$	$+0.0081$
8	0.0876	$+0.0872$	$+0.0914$	$+0.0099$

increases through $k = 4$ to $k = 8$, as described by Eq. (1). This is consistent with the mechanism, since the bound on an individual's summed contribution is tightest for individuals who contribute additional records.

The second is the saturation, which the first corpus cannot test. The registered prediction was that the gap beyond $k = 8$ would exceed the gap at $k = 8$ by less than 0.015. Across paired seeds, the increment from $k = 8$ to $k = 16$ is $+0.0004$, with an interval of $[-0.0134, +0.0141]$ ($p = 0.91$). The gap does not continue to increase, and the registered bound is met. The cost of the guarantee therefore approaches a ceiling in k on both corpora.

The asymmetry between the arms is also reproduced. Across all k, the individual-level arm varies within $[0.2116, 0.2274]$, a range of 0.0158 against the pre-specified 0.020, whereas the record-level arm increases from 0.2105 to 0.2374. As on the first corpus, the widening gap reflects improvement of the record-level arm rather than degradation of the individual-level arm. The corpora therefore differ in the magnitude of the effect but not in its mechanism, and, as registered in advance, the shape is interpreted as structural and the magnitude as corpus-dependent.

The statistical power of this replication is limited. At three seeds, no gap in Table IX is resolved: every interval covers zero, and no p-value is below 0.06. The seed standard deviation of the gap at $k = 8$ is of the same order as the gap, and resolving an effect of this size at conventional power would require approximately twenty-five seeds, against the three fixed in advance. The seed count was not increased after inspecting the results. The trend is therefore reported as directionally consistent with the VitalDB law and unresolved on this corpus. The area under the ROC curve is less informative: its gaps over the same values of k are -0.0018, -0.0021, $+0.0011$, -0.0007 and $+0.0098$, with no trend over the range covered by the first corpus. Average precision is the primary measure on this corpus, as on the first, and the area under the ROC curve provides no evidence either way.

One possible explanation for a preserved shape at a lower ceiling is offered as a hypothesis. The two tasks differ in attainable precision by approximately a factor of two: the arms on this corpus lie within $[0.1848, 0.2446]$ at a prevalence of 0.0422, whereas those on the first corpus lie between 0.4967 and 0.5955. A gap in average precision is bounded by the attainable precision, so a cost that is a fixed fraction of attainable performance would produce a smaller absolute gap on the harder task with the same ordering in k. Under this explanation, the transferable quantity is the cost relative to attainable precision rather than the ceiling A. Distinguishing this from a corpus-specific ceiling requires a third corpus.

The range of k is limited by the corpus. The registered design specified k up to 32. On this corpus, however, a budget of twenty-four windows per admission means that only $20.2\,\%$ of individuals reach thirty-two records, and these individuals have multiple admissions and are not representative of the cohort. Estimating the arm at $k = 32$ would therefore measure the effect of readmission rather than of the privacy unit. The maximum was reduced to 16, which retains $91.0\,\%$ of individuals, and $k = 32$ is reported as untested. This change was recorded as a dated amendment to the registration before any result beyond $k = 8$ was computed, and it did not alter any predicted value.

5) A matched-budget control for the external test: The comparison of the two preceding subsubsections is confounded by the training budget. The curve of Eq. (1) was *fitted* on a first-corpus grid trained at $\varepsilon_{\mathrm{train}} = 1.00$ and *tested* on a second-corpus grid trained at $\varepsilon_{\mathrm{train}} = 0.80$, the operating point of Section IV. Since additional noise reduces the difference between any two arms, which is the same direction as the observed shortfall, the shortfall cannot be attributed to the corpus without a control.

The pre-registered correction re-runs the first corpus at the operating point of the second, with k, seeds, partitions and protocol unchanged, and without refitting the forecast. Table X compares the three grids.

Taken at face value, the budget explains almost none of the shortfall. The matched gaps are not smaller than the original ones: at $k = 8$, the matched gap is $+0.0914$ against $+0.0872$, so the lower budget moved the first corpus in the *opposite* direction to that predicted by the confound. Decomposing the total change at $k = 8$ attributes $+0.0043$ to the budget and -0.0815 to the corpus.

Two qualifications apply. First, the re-run did not change the budget alone: the operating point also changes the batch size from 32 to 64 units, so the budget term is the joint contribution of the budget and the sampling configuration. Second, the matched grid fails its own positive control. At $k = 1$, where the two arms must agree by construction, it separates them by $+0.0398$, with all three seeds of the same sign, against a pre-specified tolerance of 0.020. The interval covers zero, so the result is formally unresolved, but the consistent sign is notable, and both the original grid (-0.0071) and the external

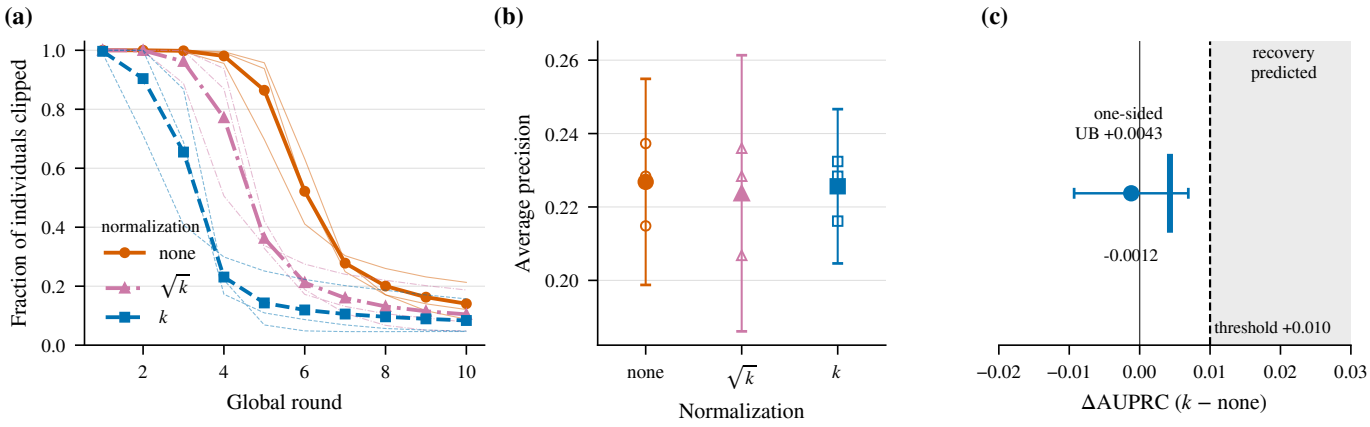


Fig. 11. Normalized clipping on the natural partition of eICU at the operating budget ($n = 3$ seeds). (a) Fraction of individuals whose summed gradient is clipped, per round; thin lines are seeds and thick lines their mean. (b) Average precision per rule, seeds and mean ± Student-t 95 % interval. (c) Paired difference between normalization by the count and the fixed bound, with its two-sided interval and one-sided 95 % upper bound (bar), against the registered recovery threshold. On one seed every rule selected the first round.

grid (-0.0011) pass the same check.

The failure is attributable to the sampling configuration rather than to the budget. At $k = 1$, the first corpus retains one record per individual, giving a training pool of 4,372 records, the smallest in this study; with a batch of 64, this yields approximately sixty-eight optimization steps per round for either arm. The original grid uses the same pool with a batch of 32 and its control passes, and the external corpus uses 76,589 records with a batch of 64, seventeen times more data at the same batch size, and its control also passes. The failing configuration combines the smallest pool with the largest batch, which indicates a lack of power rather than an effect of the privacy unit, consistent with the width of the interval.

Subtracting each grid's own $k = 1$ offset from its other rows gives a conservative estimate. At $k = 8$, the budget and sampling term then accounts for -0.0426 and the corpus for -0.0407; at $k = 4$, for -0.0284 and -0.0491. The two contributions are then comparable. Under the face-value estimate, the budget contributes nothing; under the conservative estimate, it contributes about half. Under neither does it account for the factor of nine separating the two corpora at $k = 8$, and under neither does the corpus term change sign. The matched grid therefore shows that the budget mismatch cannot account for the external shortfall, but not that the mismatch contributed nothing.

The control failure is confined to this re-run. The scaling law of Table VIII and its external test in Table IX are measured on grids whose $k = 1$ controls pass, so neither depends on the grid reported here; only the decomposition of the difference between them does. The matched grid is therefore reported as a bound on a confound rather than as a measurement.

6) Is the cost an artifact of the clipping rule?: The individual-level arm clips each individual's *summed* gradient to a fixed norm, so clipping is strongest for individuals with many records. The flatness of this arm in k could therefore result from the clipping rule rather than from the guarantee. In that case, it should be partly recoverable by dividing the summed gradient by the record count, or by its square root, before clipping. Neither normalization changes the guarantee: sensitivity is determined by the clipping norm, so the noise scale and the budget are identical under all three rules, and any difference is due to optimization.

The premise holds in direction but not in magnitude. On the external corpus, the fraction of individuals whose gradient is clipped increases from 0.148 for those with one to four records to 0.502 for those with forty-nine or more, but the rank correlation between record count and clipping is only 0.138. Normalizing by the count halves the clipped fraction, from 0.3945 to 0.2039, rather than eliminating it. The first corpus cannot test this question, because nearly every individual contributes exactly eight records, so normalization by the count is a uniform rescaling. The test is therefore performed on the external corpus only, using its natural partition rather than the capped partitions of the scaling grid, since capping would remove the variation in records per individual, which ranges from two to 192.

Normalization does not recover utility. At the operating budget and three pre-specified seeds, the individual-level arm attains an average precision of 0.2268 with the fixed bound, 0.2256 with normalization by the count, and 0.2237 with normalization by its square root. The paired differences relative to the fixed bound are -0.0012 ± 0.0081 and -0.0031 ± 0.0107. The pre-registered threshold for recovery was $+0.010$ for normalization by the count, obtained by scaling the threshold for the first corpus to the precision attainable on this corpus. The one-sided 95 % upper bound on the difference is $+0.0043$, so an effect of the registered size is excluded (Fig. 11(c)). The area under the ROC curve changes in the same direction, by -0.0017 and -0.0012. The square-root rule, predicted to fall between the other two, falls below both, although by less than the seed variation.

The intervention was effective: in every round, on every seed, normalization by the count produced the lowest clipped fraction and the fixed bound the highest, with ties only where all rules clip every individual (Fig. 11(a)). In the final round, the fractions are 0.157 against 0.213 on one seed, and 0.046 against 0.088 and 0.048 against 0.121 on the others. On one

seed, the selected checkpoint for every rule is the first round, before the rules diverge, so the three rules agree to within 10^{-4} and this seed carries little information about the contrast. On this corpus, the tested count normalization therefore does not explain the cost of the individual-level unit. On VitalDB at $k = 8$ and the same budget, count normalization lowers the final-round clipped fraction from 0.50 to 0.34 and changes AUPRC by $+0.0038$ $[-0.0072, +0.0148]$; the one-sided upper bound, $+0.0113$, does not exclude the registered $+0.010$, so this test is inconclusive by its own criterion, although even the bound is one eighth of the 0.0914 cost at that operating point. Square-root normalization changes AUPRC by less than 10^{-4}. On one VitalDB seed, the selected checkpoint is the third round, before the rules diverge.

C. Does Fusion Earn Its Place?

The benefit of combining the two modalities is evaluated separately from the federated system, because the question concerns the data rather than the protocol, and federation and privacy noise would add variance and confounds. Each arm is retrained on its own inputs rather than evaluated with one modality zeroed, since zeroing a stream of a jointly trained model measures robustness to a missing stream rather than the information it carries.

Over five seeds on the case-level partition, the fused model attains 0.8653 ± 0.0072 in area under the ROC curve, against 0.8283 ± 0.0087 for the better single modality and 0.7680 ± 0.0045 for the weaker, where each interval is a Student-t half-width. The gain of 0.0370 is 5.9 times the seed standard deviation, and the gain in average precision is 0.0671. The modalities are paired by the recording apparatus, from the same patient during the same operation, so the gain does not arise from a pairing rule. With two seeds, the same experiment gave gains of 0.0411 and 0.1062, at 14.1 times the seed standard deviation; with five seeds, the average-precision gain is $37\,\%$ lower and the effect size less than half as large. Only the five-seed values are used.

On the wearable deployment, the better single modality already attains 0.9976, leaving a fusion gain of 0.0004 and no room for a second stream to contribute. The benefit of fusion therefore depends on the task and must be established per task.

1) The external clinical corpus, and the condition that separates the three: On the external clinical corpus, which has more headroom than the wearable corpus, fusion yields no gain. The fused model attains 0.8249 in area under the ROC curve against 0.8212 for the admission-record stream alone, a gain of 0.0037, and -0.0001 in average precision. The vital-sign stream alone reaches 0.6748, so the two streams differ substantially in strength. This motivates a second hypothesis, that fusion requires the streams to be of comparable strength, which was pre-registered before testing (Fig. 13).

The test weakens the stronger stream until the two are balanced and measures whether a gain appears. Zeroing the dominant admission feature reversed the dominance rather than balancing the streams (Table XI, Fig. 12). Scaling the feature was ineffective, because the stream is standardized and its encoder is linear at the input, so scaling is compensated by the weights; a scale of 0.15 changed the stream by 0.0007. The intervention used therefore mixes the feature with fixed-seed Gaussian noise at a ratio that preserves its variance, which removes information rather than rescaling it. A calibration on the admission stream alone, with two seeds, changed its performance monotonically from 0.6252 to 0.7859 across four mixing levels, and the level closest to the vital-sign stream was selected. Because the calibration uses only one single-modality arm and never the fused model, it does not select on the outcome. Because the current partitions contain $20\,\%$ less client data than the run reported above, the comparator was re-measured on them; its gain of 0.0020 confirms that the reduction does not affect the null result.

TABLE XI
FUSION ON THE EXTERNAL CLINICAL CORPUS AS THE STRONGER STREAM IS WEAKENED. AREA UNDER THE ROC CURVE, MEAN OVER FIVE SEEDS IN EVERY ROW; GAINS ARE OVER THE BETTER SINGLE MODALITY, IN AREA UNDER THE ROC CURVE AND IN AVERAGE PRECISION (AP).

Condition	Vitals	Admission	Fused	Gain	Gain (AP)
Matched control	0.7068	0.8221	0.8241	$+0.0020$	-0.0008
Feature zeroed	0.6973	0.5947	0.6911	-0.0062	-0.0182
Titrated to parity	0.6992	0.6845	0.7282	$+\mathbf{0.0291}$	$+0.0121$

When the streams are balanced, a gain appears. With the two single-modality arms within 0.0147 of each other, the fused model attains 0.7282 ± 0.0191 against 0.6992 ± 0.0120 for the better single modality, a gain of 0.0291 ± 0.0211. The gain is positive on all five seeds, with a minimum of 0.0129 (paired t-test, $p = 0.019$; the signed-rank test gives $p = 0.0625$, its minimum attainable value at five seeds). The result exceeds the pre-registered threshold of 0.015 and is fourteen times the gain of the matched control. In average precision, the gain is 0.0121 ± 0.0215, with one negative seed, and is not resolved.

The scope of this result is limited in two respects. Balance was imposed by calibration, so the result shows that comparable stream strength is *sufficient* for a fusion gain on this corpus, but not that it is necessary. In addition, the gain is measured against a deliberately weakened stream: the balanced fused model remains well below the intact admission stream, so the result does not support fusing the two streams in the deployed configuration.

D. Necessity of the Components

Each component is removed in turn from the full system, with all other settings fixed, and the arms are paired by seed over three seeds (Table XII, Fig. 14(a)). Of the four removals, only the removal of the privacy mechanism produces an interval that excludes zero.

1) Compression is justified by traffic, not by accuracy: Removing top-κ sparsification changes the area under the ROC curve by $+0.0012$, with an interval spanning zero, and increases uplink traffic by a factor of 5.0. Removing quantization changes it by $+0.0001$, with an interval of width 0.0011, the narrowest in the study, and increases uplink traffic by a factor of 1.60. Neither component affects predictive

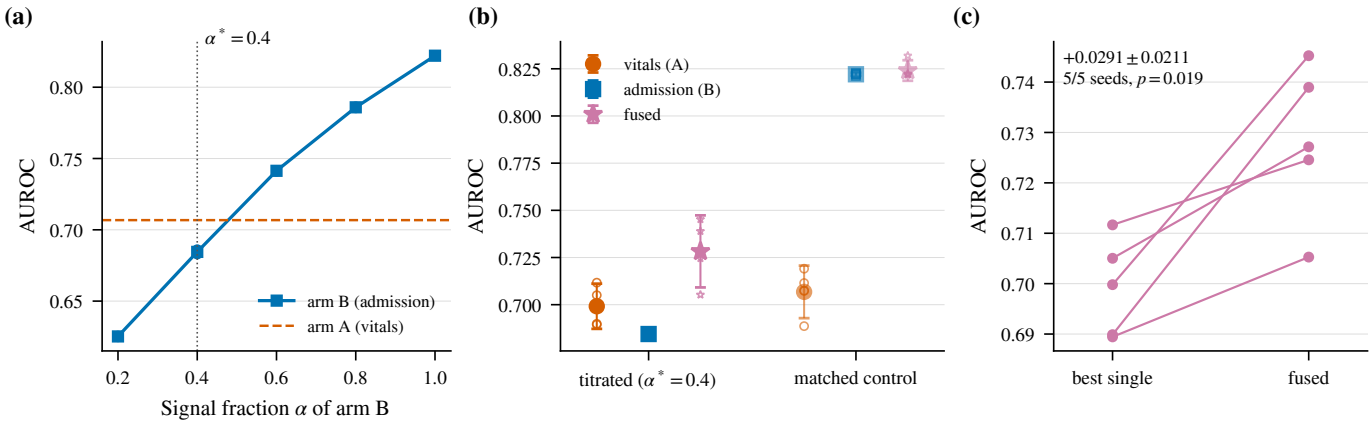


Fig. 12. Fusion-balance test on eICU. (a) Mixing the admission stream with fixed-seed noise weakens arm B monotonically; the level α^* nearest arm A is selected using arm B alone, never the fused model. (b) At parity the fused model exceeds both single modalities ($n = 5$, mean ± Student-t 95 % interval); right: the unbalanced matched control. (c) Seed-paired gain of the fused model over the better single modality; the gain in average precision, +0.0121, is not resolved.

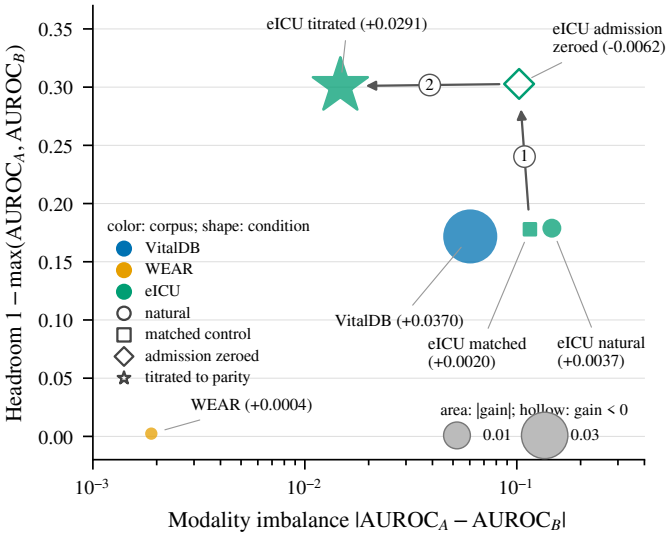


Fig. 13. Fusion gain in area under the ROC curve over the better single modality (bubble area and label) against modality imbalance and headroom. Each point is a mean over seeds (five; two for WEAR). Filled: positive gain; hollow: negative. Color: corpus; marker shape: condition. Numbered arrows trace the eICU titration from the matched control.

TABLE XII
COMPONENT ABLATION ON THE CLINICAL DEPLOYMENT (CASE-LEVEL PARTITION), UNDER THE RECORD-LEVEL MECHANISM. ΔAUROC IS PAIRED AGAINST THE FULL SYSTEM BY SEED OVER THREE SEEDS, REPORTED AS MEAN ± THE HALF-WIDTH OF A STUDENT-t INTERVAL; AN INTERVAL EXCLUDES ZERO WHEN THE HALF-WIDTH IS SMALLER THAN THE MEAN, AND THE ONE THAT DOES IS SET IN BOLD. MB IS THE TOTAL COMPRESSED PAYLOAD UPLOADED ACROSS TRAINING. C6 IS AN *inclusion* ARM: FOR IT A RISE, NOT A FALL, WOULD INDICATE A CONTRIBUTION.

Removed	AUROC	ΔAUROC	MB
— (full)	0.8541	—	252
C1 hierarchy	0.8484	-0.0058 ± 0.0088	50
C2 privacy	0.8307	$\mathbf{-0.0234 \pm 0.0205}$	252
C3 sparsification	0.8553	$+0.0012 \pm 0.0083$	1260
C4 quantization	0.8543	$+0.0001 \pm 0.0006$	403
+C6 ASWA	0.8542	$+0.0001 \pm 0.0009$	252

performance; together they reduce the payload eightfold at no measurable cost in discrimination.

2) The edge tier does not cost accuracy: Removing the edge tier changes the area under the ROC curve by -0.0058 at three seeds, with an interval covering zero. With the six seeds specified in the pre-registration, the effect is resolved on both deployments: -0.0047 $[-0.0088, -0.0006]$ ($p = 0.032$) on the clinical corpus and -0.0052 $[-0.0074, -0.0030]$ ($p = 0.0018$) on the wearable corpus. Removing the tier therefore slightly reduces performance.

This result is interpreted as a bound on the cost of the hierarchy rather than as a benefit, because the flat arm differs in two respects. It uses a single edge with one edge round per global round, whereas the hierarchical arm uses five, so it changes both the aggregation schedule and the topology. The small advantage may be due to the additional aggregation steps rather than to the tier itself; separating the two would require a flat arm matched on aggregation steps. For the same reason, the uplink value of this arm is not comparable with the others. The comparison shows that the edge tier does not reduce predictive utility, so the communication and energy savings of Section V-A are not obtained at the expense of accuracy.

3) Removing the privacy mechanism does not recover utility: Training the same system without differential privacy *reduces* the area under the ROC curve by 0.0234 $[-0.0439, -0.0029]$ and increases the seed-to-seed spread in average precision from ± 0.014 to ± 0.070. Two of the three non-private seeds reach their best validation score in the first of ten rounds and degrade thereafter, whereas every private seed peaks between rounds six and nine. Clipping and calibrated noise therefore act as a regularizer for a compact head trained over frozen features, and without them the head overfits early.

This comparison is against a baseline without additional regularization beyond checkpoint selection, not against a non-private system tuned for generalization, which would typically use weight decay or early stopping. The result shows that removing the guarantee does not recover utility on this task; it does not show that differential privacy improves accuracy in

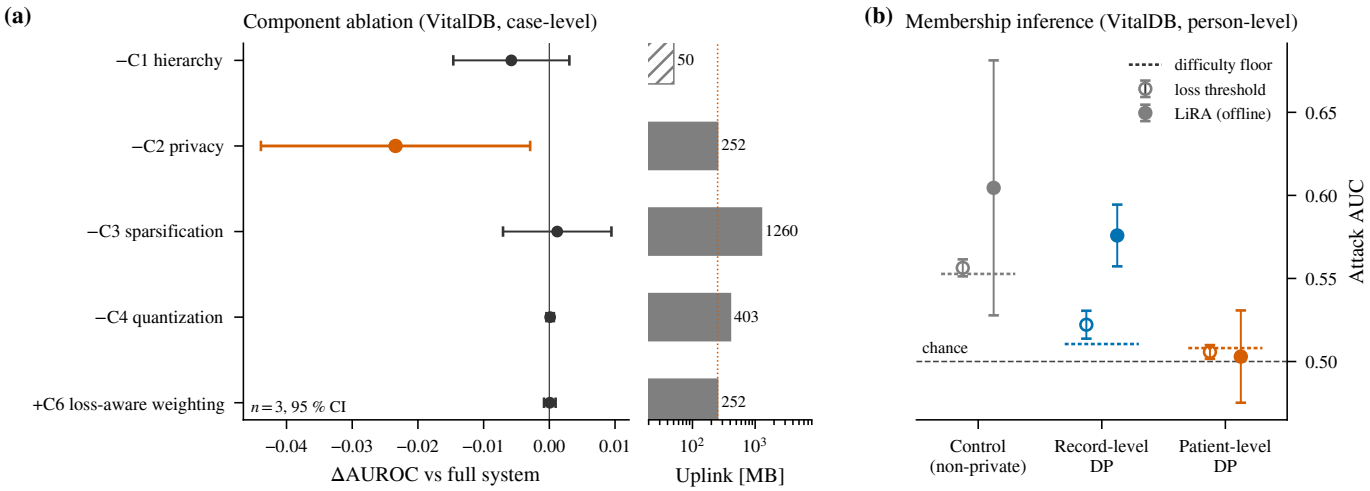


Fig. 14. (a) Component ablation on VitalDB (case-level partition): change in area under the ROC curve against the full system, paired by seed ($n = 3$, Student-t 95 % interval), beside each arm's total uplink. Only removing the privacy mechanism resolves. The hatched bar is not comparable, because removing the hierarchy changes the schedule. (b) Membership inference (person-level) with sixteen shadow models, 600 members and 600 non-members: mean attack AUC ± Student-t 95 % interval over seeds for the loss-threshold (open) and likelihood-ratio (filled) attacks; dotted: each configuration's difficulty floor.

general. It also shows that the accuracy cost of the guarantee, bounded at two percentage points in Table V, is not incurred at this operating point.

4) Loss-aware edge weighting: The aggregation rule ASWA, inherited from an earlier version of this framework, weights each edge at the cloud by both its data volume and the mean local loss of its clients, instead of by data volume alone. It was evaluated as an inclusion arm. Adding it to the full system changes the area under the ROC curve by $+0.0001$, with an interval of width 0.0018 that covers zero, and leaves uplink traffic unchanged. Since this is an inclusion arm, an increase would indicate a contribution, and none is observed. The component is therefore not adopted.

TABLE XIII
MEMBERSHIP INFERENCE ON THE PERSON-LEVEL VITALDB PARTITION. MEAN ATTACK AUC OVER SEEDS WITH 95 % INTERVALS. THE FLOOR IS THE AUC OBTAINED FROM PER-PATIENT DIFFICULTY ALONE, ESTIMATED FROM THE SHADOWS WITHOUT THE TARGET MODEL. LIRA IS THE OFFLINE LIKELIHOOD-RATIO ATTACK. THE CONTROL IS NON-PRIVATE TRAINING OF THE SAME HEAD OVER THE SAME FROZEN ENCODERS (THREE SEEDS; FIVE OTHERWISE).

Configuration	Loss AUC	LiRA AUC	Floor
Control (no DP)	0.556 [0.551, 0.561]	0.605 [0.528, 0.681]	0.553
Record-level DP	0.522 [0.514, 0.531]	0.576 [0.557, 0.594]	0.511
Patient-level DP	0.506 [0.502, 0.510]	0.503 [0.475, 0.531]	0.508

E. Membership-Inference Audit

Table XIII and Fig. 14(b) report the empirical leakage of three configurations on the person-level VitalDB partition under the attacks of Section III-K. Each attack uses sixteen shadow models trained on independently sampled halves of the patient population, and each configuration is scored on the same 600 member and 600 non-member patients. The audited record- and patient-level models are the five-seed models of the unit comparison (Section V-B), trained at $\varepsilon = 1.0$ with $\varepsilon_{\text{conf}} = 0.5$ and one worst-client multiplier under the Rényi accountant, a slightly weaker budget than the headline $\varepsilon = 0.9$.

Attack AUCs are compared with a difficulty floor rather than with 0.5. Members are drawn from the training patients and non-members from the test patients, so the two groups can differ in intrinsic difficulty, and an attack can exceed 0.5 by detecting difficulty rather than membership. The floor ranks patients by their mean statistic over the shadows that exclude them, without using the target model.

1) Positive control: On non-private training of the same head, the offline likelihood-ratio attack reaches 0.605, 0.052 above the floor of 0.553 and 0.048 above loss thresholding, which per-patient calibration outperforms because a global threshold is confounded by difficulty (Fig. 4). Its interval spans the floor at three seeds; the online attack with a global variance, 0.597 $[0.586, 0.609]$, lies entirely above it. The audit therefore detects membership beyond difficulty, but through a different attack from the one that separates the two privacy units below. The likelihood-ratio attack assumes that the shadows reproduce the scale of the target; regressing each target's per-patient statistic on the mean of its OUT shadows gives slopes of 1.07–1.16 for the control, 0.95–0.99 for the record-level arm and 0.93–1.25 for the patient-level arm, close to the ideal of one.

2) Record- against patient-level privacy: Record-level training remains detectably above its floor under offline LiRA (0.576, margin $+0.065$), whereas patient-level training does not (0.503, margin -0.005). The record-level arm is not equivalent to the control: its offline margin is similar ($+0.065$ against $+0.052$), but it lies below the control under the loss threshold (0.522 against 0.556) and the online attack with a global variance (0.508 against 0.597), and the two were not compared statistically (three control seeds against five). The patient-level arm lies at its floor, with offline AUCs of 0.479–0.527 across seeds. Paired over five seeds on identical shadow splits, the record-level arm exceeds the patient-level arm by

$+0.073$ $[0.054, 0.092]$ ($p < 0.001$) for the offline likelihood ratio and by $+0.016$ $[0.010, 0.023]$ ($p = 0.002$) for the loss threshold; both survive a Bonferroni correction over the four attacks. The two online variants, -0.010 $[-0.037, +0.016]$ and $+0.016$ $[-0.022, +0.053]$, resolve no difference. True-positive rates at a $1\,\%$ false-positive rate are 0.024 for the control and 0.017 for both private arms; at $0.1\,\%$ the rate is undefined, because 600 non-members cannot resolve it.

The direction of the prediction is confirmed, but not its magnitude: the record-level attack reaches 0.576, not the predicted 0.7. On the case-level partition, where members were operations, the same comparison resolved no difference ($+0.029$ $[-0.038, +0.095]$ for the offline likelihood ratio). The audit is thus consistent with the analytic argument of Section III-E: under the offline attack a record-level budget leaves a measurable patient-level signal, whereas the patient-level guarantee leaves none. The audit covers VitalDB only and bounds what these four attacks achieve; it does not establish the absence of leakage.

F. Attribution Under the Privacy Mechanism

Integrated gradients attribute the positive-class prediction of the clinical deployment (case-level partition) to its 52 inputs over 512 test windows, with 64 integration steps from a mean baseline. Of these inputs, 22 indicate whether a measurement was taken rather than its value. The share of total attribution carried by these indicators is $19.9\,\%$ when the full model is fine-tuned without privacy and $28.2\,\%$ under record-level differential privacy at $\varepsilon = 1$, a seed-paired increase of 8.4 points (from 4.6 to 11.0 per seed, $p = 0.050$, three seeds). With the encoders frozen, the share is $15.8\,\%$ without privacy, $21.6\,\%$ under the record-level guarantee and $21.7\,\%$ under the patient-level guarantee (five seeds). On the three paired seeds the patient-level share is higher by 1.4 points (from 1.3 to 1.5 per seed, $p = 0.002$): the difference is resolved but small next to the 8.4-point effect of fine-tuning under privacy.

The indicators make up $42\,\%$ of the inputs, so in no configuration do they carry more than their proportion; the measured effect is the increase under privacy, not a dominance of missingness. The ranking of individual inputs is not stable: the five highest-ranked inputs change between seeds in every configuration, although the mean rank correlation between seeds is 0.84–0.99. No individual input is therefore identified as decisive. Attribution measures what the trained model uses, not what its accuracy depends on.

G. Comparison with Prior Systems

Two prior systems were re-implemented and trained through the same trainer, splits, frozen encoders, number of rounds and five seeds as PP-HFM (Table XIV). HED-FL [3] aggregates hierarchically without a privacy mechanism, uploads dense updates and samples $60\,\%$ of the clients in each round. FL2DP [9] perturbs every coordinate of the update with an exponential mechanism and shuffles the perturbed updates, without an edge tier. FL2DP was run on the first clinical corpus only, because its per-coordinate mechanism would require about two GPU days per seed on the external one.

TABLE XIV

PP-HFM AGAINST TWO RE-IMPLEMENTED PRIOR SYSTEMS, FIVE SEEDS PAIRED. ε IS THE PATIENT-LEVEL TRAINING BUDGET AT $\delta = 10^{-5}$; A DASH DENOTES A MECHANISM WITHOUT A FINITE BOUND. MB IS THE TOTAL UPLINK. LIRA IS THE OFFLINE ATTACK AUC, WITH THE MARGIN ABOVE THE PER-CONFIGURATION DIFFICULTY FLOOR IN PARENTHESES; IT WAS NOT RUN ON EICU. THE PP-HFM LIRA ENTRY IS THE PATIENT-LEVEL AUDIT TARGET ($\varepsilon = 1.0$, $\varepsilon_{\mathrm{conf}} = 0.5$, ONE WORST-CLIENT MULTIPLIER, RÉNYI ACCOUNTANT), NOT THE $\varepsilon = 0.9$ MODEL WHOSE AUROC AND UPLINK ARE LISTED. EVERY AUROC DIFFERENCE FROM PP-HFM EXCLUDES ZERO EXCEPT HED-FL ON VITALDB.

	System	ε	AUROC	MB	LiRA
VitalDB	PP-HFM	0.90	**0.8318**	**252**	**0.503** (-0.005)
	HED-FL	∞	0.8262	560	0.581 ($+0.026$)
	FL2DP	—	0.8195	1008	0.608 ($+0.049$)
eICU	PP-HFM	0.90	0.7936	**5235**	—
	HED-FL	∞	**0.8252**	12541	—

1) Utility and traffic: On VitalDB the private PP-HFM model exceeds FL2DP by 0.0123 in area under the ROC curve $[0.0064, 0.0181]$ and by 0.025 in average precision $[0.015, 0.035]$, and HED-FL by 0.030 in average precision $[0.005, 0.055]$; the difference from HED-FL in area under the ROC curve, 0.0056 $[-0.0018, +0.0131]$, is not resolved. HED-FL exceeds the non-private reference of PP-HFM by 0.0062 $[0.0011, 0.0113]$. On the external corpus HED-FL exceeds the private model by 3.17 points of area under the ROC curve $[1.56, 4.77]$ and 1.77 points of average precision $[0.75, 2.78]$, and is level with the non-private reference ($+0.0008$, $p = 0.39$); the gap is therefore the price of the patient-level guarantee measured in Section V-A3. PP-HFM uploads 2.2 times less than HED-FL on VitalDB and 2.4 times less on eICU, because only the sparsified, quantized head update leaves the client, and 4.0 times less than FL2DP.

2) Leakage: On VitalDB, both baselines are attacked with the audit of Section V-E, on the same seeds, shadow splits, members and non-members as the PP-HFM arm. Offline LiRA reaches 0.581 against HED-FL and 0.608 against FL2DP, 0.078 $[0.050, 0.106]$ and 0.105 $[0.079, 0.130]$ above PP-HFM; the loss threshold gives $+0.047$ $[0.040, 0.053]$ and $+0.055$ $[0.051, 0.059]$. Both baselines also have higher difficulty floors (0.555 and 0.559 against 0.508), and their margins above them, the signal specific to the target model, are $+0.026$ and $+0.049$ against -0.005 for PP-HFM. The PP-HFM is the patient-level audit target, trained at $\varepsilon = 1.0$ with $\varepsilon_{\mathrm{conf}} = 0.5$ and one worst-client multiplier under the Rényi accountant, not the $\varepsilon = 0.9$ model whose AUROC and uplink are listed, so its leakage is measured at a slightly weaker budget than the headline. The comparison bounds what these attacks achieve and does not by itself establish the guarantee. FL2DP thus leaks more than HED-FL, which adds no noise at all. Its candidate set is constructed from the private gradient value (Eqs. 11–12 of [9]), so the mechanism satisfies no finite ε, and the audit shows that its perturbation does not reduce leakage either.

VI. Discussion

A. What the Price of the Unit Depends On

The central finding of this paper is not that an individual-level guarantee has a cost, which was expected, but that the cost is not constant. At a single operating point, the guarantee reduces the area under the ROC curve by 0.0170. When the per-step budget and the number of units per step are held fixed, so that only the clipping geometry differs, the penalty is indistinguishable from zero at one record per individual and increases monotonically with the number of records, reaching 0.0872 in average precision at eight records and largely saturating before that point.

Two features of this measurement are important. First, the design contains its own control: at one record per individual, the two units coincide by construction, so the arms must agree, and they do (-0.0071, with an interval covering zero). The divergence at larger contributions can therefore be attributed to the unit. Second, the divergence is asymmetric. The individual-level arm remains flat across the whole range, whereas the record-level arm improves. The widening gap thus reflects the absence of improvement under the stricter unit rather than degradation, and an individual-level guarantee becomes relatively more expensive, not cheaper, as more records are collected per patient.

The external corpus supports the shape of this result but not its magnitude. On a second clinical corpus with real hospitals as clients, a median of 444 individuals per client and encoders pretrained on a verified disjoint population, the ordering is reproduced, from a negative gap at $k = 1$ to an increasing positive gap through $k = 8$. The registered saturation is also reproduced: from $k = 8$ to $k = 16$, the gap changes by $+0.0004$ $[-0.0134, +0.0141]$, so the cost approaches a bound in k. The magnitude, however, is approximately one ninth of the pre-registered forecast and is not resolved at three seeds. The shape of the law is therefore interpreted as structural and its scale as corpus-dependent, and the two corpora are not combined into a single estimate.

One possible explanation is that the two tasks differ in attainable precision by approximately a factor of two. Because a gap in average precision is bounded by the attainable precision, a cost that is a fixed *fraction* of attainable performance would produce a smaller absolute gap on the harder task with the same ordering in k. Under this explanation, the transferable quantity is the cost relative to attainable precision rather than the ceiling. A third corpus is required to distinguish this explanation from a corpus-specific ceiling.

The comparison between corpora is subject to a budget confound, which is bounded but not eliminated. The first corpus was originally trained at a higher budget, and additional noise reduces differences between arms. At the matched operating point, the gaps on the first corpus are not smaller, so at face value the budget explains none of the shortfall. The matched grid, however, fails its own $k = 1$ control, and under the conservative assumption that this offset applies at every k, the budget accounts for approximately half of the difference at $k = 8$. Under neither interpretation does it account for the factor of nine, and under neither does the corpus term change sign.

The tested count normalization does not explain the cost on eICU (Section V-B6). Clipping the summed contribution affects individuals with many records most strongly, but normalizing the clipping bound by the record count, which reduced the clipped fraction in every round, did not improve utility, and the registered effect size is excluded by the one-sided upper bound ($+0.0043$). This conclusion rests on three seeds, one of which selected a first-round checkpoint. On VitalDB, where records per individual are capped at k, the same test is inconclusive (upper bound $+0.0113$), but even that bound is one eighth of the cost, so the clipping rule accounts for at most a small part of it.

B. What the Initialization Provides, and What the Privatized Subset Does Not

The proposed recipe differs from conventional private training in two respects: it starts from pretrained encoders and privatizes only the fusion head. Compared with full-model private training from a random initialization, it shows a large advantage that grows as the budget tightens. The three-arm decomposition shows that this advantage is due almost entirely to the initialization. Under privacy, restricting the privatized subset changes the result by at most 0.0059 on the clinical corpus and 0.0013 on the wearable corpus, with intervals covering zero. Across the full budget range, the randomly initialized arm loses 0.1906 and 0.3716 on the two corpora, while the pretrained head-only arm loses 0.0225 and 0.0157; on the wearable corpus, the randomly initialized arm falls to chance.

Two consequences follow. First, robustness to a tight budget depends on the initialization rather than on the fraction of the network that is privatized, so obtaining a legitimate pretrained initialization should take priority over selecting the privatized parameters. Second, head-only training cannot be justified by a reduction in noise. The noise scale under a given budget depends on the budget, the number of steps and the sampling rate, not on the number of parameters, so freezing most of the network does not reduce the noise per remaining parameter. Head-only training instead reduces communication, since only head parameters are transmitted, which yields the uplink reductions of $90.8\,\%$ and $96.1\,\%$ at no measurable cost in accuracy.

C. Which Components Are Necessary

In the component ablation, only the removal of the privacy mechanism produces an interval that excludes zero. The compression components do not affect accuracy: removing sparsification changes it by $+0.0012$ and increases uplink traffic fivefold, and removing quantization changes it by $+0.0001$ and increases the payload by three fifths. These well-powered null results support the use of both components for traffic reduction.

Removing the edge tier slightly reduces accuracy: the effect is not resolved at three seeds (-0.0058) but is resolved at the six pre-specified seeds, at -0.0047 $[-0.0088, -0.0006]$

on the clinical corpus and -0.0052 $[-0.0074, -0.0030]$ on the wearable corpus. This is not interpreted as evidence that hierarchical aggregation improves discrimination, because the flat arm performs one edge round per global round instead of five and therefore changes the aggregation schedule together with the topology. A flat arm matched on aggregation steps would be required to separate the two. The result establishes that the edge tier does not reduce predictive utility, so the uplink and energy savings are obtained without an accuracy penalty.

Removing the privacy mechanism *reduces* accuracy by 0.0234 and increases the seed-to-seed spread in average precision fivefold. Two of three non-private seeds peak in the first of ten rounds and then degrade, whereas all private seeds peak between rounds six and nine, which indicates that clipping and noise regularize a compact head over frozen features. This applies to a small head on frozen encoders and does not imply that differential privacy is without cost; the privacy-unit results quantify that cost under a different comparison. The tested inclusion arm was not adopted, as it produced no measurable effect.

D. Position Against Prior Art

PP-HFM builds on two established mechanisms, user-level differential privacy and parameter-efficient private fine-tuning. Its contribution is their composition inside a hierarchy, with a proof of the resulting guarantee, and the measurement of its cost. Hierarchical federated learning has been developed for energy and communication efficiency without a formal privacy guarantee, and private federated learning without a hierarchy, so the cost of including the edge tier in the privacy accounting has not been measured. Nor has the appropriate unit of the guarantee been examined for settings in which each individual contributes many records, which is the typical case in clinical and wearable sensing but uncommon in the benchmarks used to evaluate private federated learning.

The re-implemented baselines of Section V-G place PP-HFM against both lines. On the external corpus HED-FL, the hierarchy without privacy, matches the non-private reference of PP-HFM, so its 3.2-point lead in AUROC is the price of the patient-level guarantee. On VitalDB, PP-HFM matches HED-FL in AUROC, exceeds it in AUPRC, and leaks less under offline LiRA (margin -0.005 against $+0.026$). FL2DP adds noise yet leaks more than HED-FL, which adds none. Its candidate set is built from the private gradient value, so it satisfies no finite ε: a perturbation without a valid bound is not a weaker guarantee but no guarantee, and only an audit can show what it protects.

The contributions are therefore measurements and the auditing practice under which they are reported, and several are negative: the privatized subset contributes nothing once initialization is controlled, compression does not affect accuracy, and the tested inclusion arm provides no benefit. Each is reported with the control or power analysis required to interpret it. The membership-inference result is positive on VitalDB: with a control that detects leakage, record-level training remains detectably above its difficulty floor under offline LiRA and patient-level training does not, although the difference is smaller than predicted and is resolved by two of four attacks.

E. Practical Implications

Four recommendations follow from the measurements.

First, the unit of the guarantee should be chosen explicitly and its cost evaluated as a function of the number of records per individual. In settings where each individual contributes many records, a record-level budget does not bound the relevant quantity, and a post hoc group-privacy conversion is prohibitively expensive, whereas direct individual-level accounting costs noise rather than budget. The cost is predictable from the structure of the data and is bounded in the number of records per individual.

Second, a legitimate pretrained initialization should be obtained before other aspects of the private recipe are tuned. Under a tight budget, it has the largest effect of any decision measured here, and the disjointness condition required for an end-to-end guarantee has a measured and moderate cost.

Third, compression should be used. Both compression components have no measurable effect on accuracy and account for most of the uplink reduction, and the edge tier provides further communication and energy savings without an accuracy penalty.

Fourth, the benefit of fusion should be established per task rather than assumed from the availability of two sensors. The gain is 0.0370 in area under the ROC curve on the first clinical corpus, cannot be evaluated on the wearable corpus, whose better single modality already reaches 0.9976, and is absent on the external clinical corpus unless the two streams are balanced (Section V-C1).

F. Limitations

The privacy-unit results rest on two clinical corpora and cannot be tested on the third. In the wearable corpus, each client corresponds to one participant, so client-level, user-level and individual-level guarantees coincide and the divergence addressed by the mechanism cannot occur. The external clinical corpus reproduces the ordering and the saturation of the scaling law, but not its magnitude, which is approximately one ninth of that of the first corpus and unresolved at three seeds. The two corpora show that the law is not specific to one cohort, but they do not determine its constant, and no constant is reported.

The fusion gain is established on one corpus and, under imposed balance, on a second. The wearable corpus cannot test fusion, and on the external clinical corpus fusion yields no gain in its natural configuration. The titration shows that balance between the streams is sufficient for a gain in area under the ROC curve on that corpus, but not that it is necessary, and the gain is measured against a deliberately weakened stream. The average precision gain is not resolved. No unconditional fusion benefit is claimed.

Utilities on the two original corpora are obtained with the oracle encoder configuration. As stated in Section IV-B, the encoders for these results were pretrained on the pooled

client training data, which does not satisfy assumption (A2). The configuration with disjoint encoders costs 0.0157 in area under the ROC curve on VitalDB, so the absolute utilities on these corpora are upper bounds. Differences between arms, on which the claims rely, are unaffected because all arms share the same encoders. This limitation does *not* apply to the external clinical corpus, whose encoders were pretrained on $526{,}867$ windows verified to be disjoint from every client's training data.

The leakage difference is resolved by two of four attacks. The offline likelihood ratio and the loss threshold resolve the difference between the units; the two online variants do not. The control is resolved above its floor by the online attack with a global variance, while the offline interval spans the floor at three seeds, so the attack that validates the control is not the one that separates the units. The control (three seeds) was not compared statistically with the record-level arm (five seeds), and the record-level arm lies below it under two attacks, so the results establish no equivalence between them. The record-level attack AUC, 0.576, is well below the predicted 0.7, and an empirical audit bounds what these attacks achieve without establishing the absence of leakage.

The audit uses sixteen shadow models. The per-individual in- and out-distributions are estimated from approximately eight observations each, which is why a pooled-variance variant is also reported. A larger shadow population would improve these estimates. In addition, 600 non-members are insufficient to resolve a $0.1\,\%$ false-positive rate, so the low-false-positive regime is characterized only at $1\,\%$. The audit is performed on VitalDB only. On eICU, sixteen shadow models per arm would require approximately 160 hours of training at the cost of one run reported here, and on WEAR each client holds a single individual, so the two units do not differ.

The utility objective is not met everywhere. On the external deployment the guarantee costs 3.09 points of area under the ROC curve $[1.37, 4.81]$, an interval that spans the 2-point requirement although the point estimate exceeds it, and 3.91 $[2.40, 5.42]$ end to end, entirely above it; balanced accuracy meets it only with training-fitted constants; the cost is set by the smallest hospitals, whose noise multipliers are the largest. On the wearable deployment the accuracy cost is 1.967 pp with an interval of $[1.760, 2.173]$, which spans the requirement. The seed counts were fixed in advance and were not increased.

Several supporting measurements use few seeds. Each point of the privacy-budget sweep in Fig. 8 uses three seeds. This is sufficient for the initialization effect, which is much larger than the seed spread, but the near-zero difference between the two pretrained arms is bounded rather than measured, and the supported conclusion is that restricting the privatized subset has no detectable effect at this power. The aggregation-rule comparison that led to a component being excluded was based on one seed.

Two VitalDB partitions. Only the privacy cost, the cost of the unit, the baseline comparison and the audit are person-level. The remaining VitalDB results are case-level; for the $3.9\,\%$ of persons with several operations, their guarantee extends to the person only by group privacy.

The prior systems are re-implementations. HED-FL and FL2DP are implemented from their papers inside the shared trainer, so implementation details their authors did not report may differ. FL2DP runs on VitalDB only, because its per-coordinate mechanism would need about two GPU days per seed on eICU, and the audit of both baselines is likewise limited to VitalDB.

Only the network is simulated at packet level. Communication delay, delivery reliability and traffic volume are measured in a packet-level simulation, but the compute terms of the round latency (sensing, preprocessing, inference and aggregation) are modeled analytically and are identical across topologies. Comparisons between topologies are therefore valid, but the absolute round latency is not a measurement of a deployed system.

The interpretability evidence is associational. The attribution analysis (Section V-F) shows that the share of attribution carried by indicators of whether a measurement was taken increases under the privacy mechanism, by 8.4 points on the clinical corpus. It does not show that accuracy depends on these indicators, which would require removing them.

VII. Conclusion

This paper addressed a mismatch between what private hierarchical federated learning guarantees and what clinical IoT requires: existing systems protect individual records, whereas a clinical deployment must protect patients, each of whom contributes many records. It presented a hierarchical federated learning system for multimodal clinical and wearable sensing that enforces differential privacy at the level of the individual rather than the record, and quantified the cost of that choice. On three deployments the training budget is 0.900 on both clinical deployments and 0.766 on the wearable deployment, where the guarantee is record-level; the guarantee is proved for the training transcript and holds end to end only when the encoders and standardization constants are fitted on public data, which among the evaluated configurations is the external clinical one, and there for the $97.1\,\%$ of training individuals treated at a single hospital. The deployments therefore carry different evidence: the external corpus tests the patient-level guarantee end to end for individuals at one hospital, the first clinical corpus tests it on the training transcript, and the wearable corpus evaluates the system at record level. The guarantee costs no utility on the first clinical deployment, partitioned by person, and 3.1 points of area under the ROC curve $[1.4, 4.8]$ on the external one, inconclusive against the two-point target, and 3.9 $[2.4, 5.4]$ end to end, which misses it. Uplink traffic is reduced by $90.8\,\%$, $96.1\,\%$ and $92.3\,\%$, modeled energy per round by $72.3\,\%$, $76.8\,\%$ and $79.6\,\%$, and the 95th-percentile inference latency on a workstation CPU is 2.3, 0.75 and 0.24 ms against a requirement of 100 ms. At 187 clients on two edges, update delivery reaches $97.83\,\%$, below the $99\,\%$ target, because contention on the shared medium limits every configuration; with eight edges the target is met in every round. Because noise is added to gradients during training, the privacy mechanism adds no computation at inference.

Two findings are central. The first is that, on the clinical and wearable corpora, robustness to a tight privacy budget is determined by the initialization rather than by the size of the privatized subset. A pretrained initialization improves on a random one by 0.0129 without noise and by 0.1991 at the tightest budget, whereas under privacy restricting the privatized subset changes the result by at most 0.0059, with intervals covering zero, despite privatizing 3.3 times fewer parameters. The restriction is therefore justified by its communication savings only.

The second is that the cost of an individual-level guarantee increases with the number of records each individual contributes. At a fixed per-step budget and number of units per step, the penalty is indistinguishable from zero at one record per individual and reaches 0.0872 in average precision at eight, following the one-parameter shape $A(1-1/k)$ predicted by Proposition 1. Because the two units coincide at one record per individual, this configuration serves as a built-in control. The individual-level arm remains flat while the record-level arm improves, so the cost arises from additional records that the stricter unit cannot exploit. On the external corpus, normalizing the clipping bound by the record count does not reduce this cost. On a second clinical corpus of $132{,}736$ individuals across 187 hospitals, the ordering and the saturation transfer, whereas the magnitude is approximately one ninth of the pre-registered forecast and is not resolved at three seeds.

On VitalDB, a patient-level audit with a positive control is consistent with the unit mattering: record-level training remains detectably above its difficulty floor under offline LiRA (AUC 0.576), whereas patient-level training does not (0.503), a difference of 0.073 $[0.054, 0.092]$. Two of four attacks resolve the difference, and the audit bounds these attacks without establishing the absence of leakage.

Several results are negative. Compression does not affect accuracy and is justified by traffic reduction alone. The benefit of multimodal fusion is conditional: it is 0.0370 in area under the ROC curve on the first clinical corpus, untestable on the wearable corpus, where the better single modality reaches 0.9976, and absent on the external clinical corpus. A titration on the external corpus shows that balance between the two streams is a sufficient condition: when the stronger stream is degraded to parity with the weaker, fusion yields a significant gain in area under the ROC curve over the better single modality, although not in average precision and not relative to the intact stronger stream.

Several directions remain open. First, a third corpus is needed to establish whether the saturation level of the scaling law is a constant of the guarantee or a fixed fraction of the precision attainable on a task. Second, on eICU the cost is set by the smallest hospitals, whose high sampling rates force the largest noise multipliers; sampling patients across the clients of an edge is the most direct route to meeting the utility target there. Third, clipping each patient's gradient around its own mean, which the analysis indicates could recover part of the unit's cost, should be tested. Finally, the audit should be extended to eICU and, with more than sixteen shadow models and a larger non-member set, to the low-false-positive regime.

Data Availability

VitalDB [24] (https://vitaldb.net) and WEAR [25] are publicly available. The eICU Collaborative Research Database [36] is available through PhysioNet [37] to credentialed users under a data use agreement, so no patient-level records or arrays derived from them are redistributed. All corpora are de-identified; this study is a secondary analysis with no patient contact. Code and aggregate results will be released upon publication.

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